Heart, Lung and Critical Care Clinical Group

NHS
Guy's and St Thomas'
NHS Foundation Trust

Royal Brompton Hospital
Harefield Hospital



PT4000/PT7003 REVISION SESSION

LONDON METROPOLITAN UNIVERSITY

BEFORE WE GET STARTED…

HOW MUCH DO YOU REMEMBER?



Q1: Which of the following 2 statements describe EXTERNAL respiration?

The movement of gas in to and out of alveoli

The exchange of gases between alveoli and pulmonary capillaries

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Q2: The rate of ventilation is regulated in order to…

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2

Increase interpulmonary pressure

3

Ensure that the respiratory muscles are not overworked

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Part of the respiratory system which is no longer functional

Part of the respiratory system that does not take part in gaseous exchange

The diffusion membrane of the alveoli is not functioning

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Q4: Which of the following are functional components of the respiratory tract?

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B) The airway surface liquid layer

C) The cilia on the surface of ciliated cells.

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- Diaphragm
- Internal intercostal muscles
- External intercostal muscles
- Scalenes

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Q6: What does ‘tidal volume’ mean?

The maximum amount of air that can fill the lungs

The total amount of air that can be expired after fully inhaling

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The total amount of air that can be expired after fully inhaling

The volume of air entering or leaving the lungs during a single breath ✅

Q7: Alveolar ventilation is more important than pulmonary ventilation…?

TRUE

FALSE

Q7: Alveolar ventilation is more important than pulmonary ventilation…?

TRUE ✅

FALSE


I DON'T KNOW WHATS GOING ON
AND AT THIS POINT I'M TOO AFRAID TO ASK
memegenerator.net

Definitions:

'We breathe to ventilate and we ventilate to respire'
Tobin 1991

BREATHING:

1. The process of inhalation and exhalation
2. The process by which the 'respiratory pump' creates ventilation

VENTILATION:

Gas movement between the outside and the alveoli

RESPIRATION:

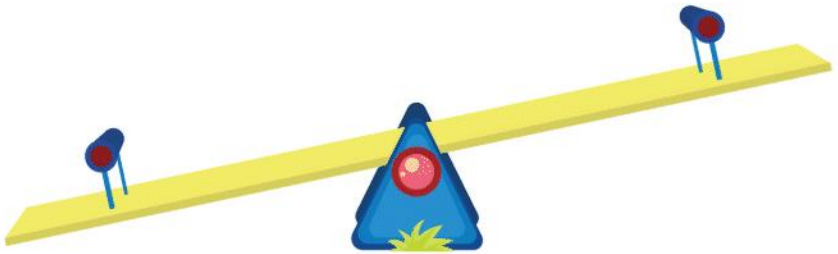
Exchange of gases between the environment and tissue cells, by:

- External respiration > alveolar-capillary level
- Internal respiration > capillary-tissue level

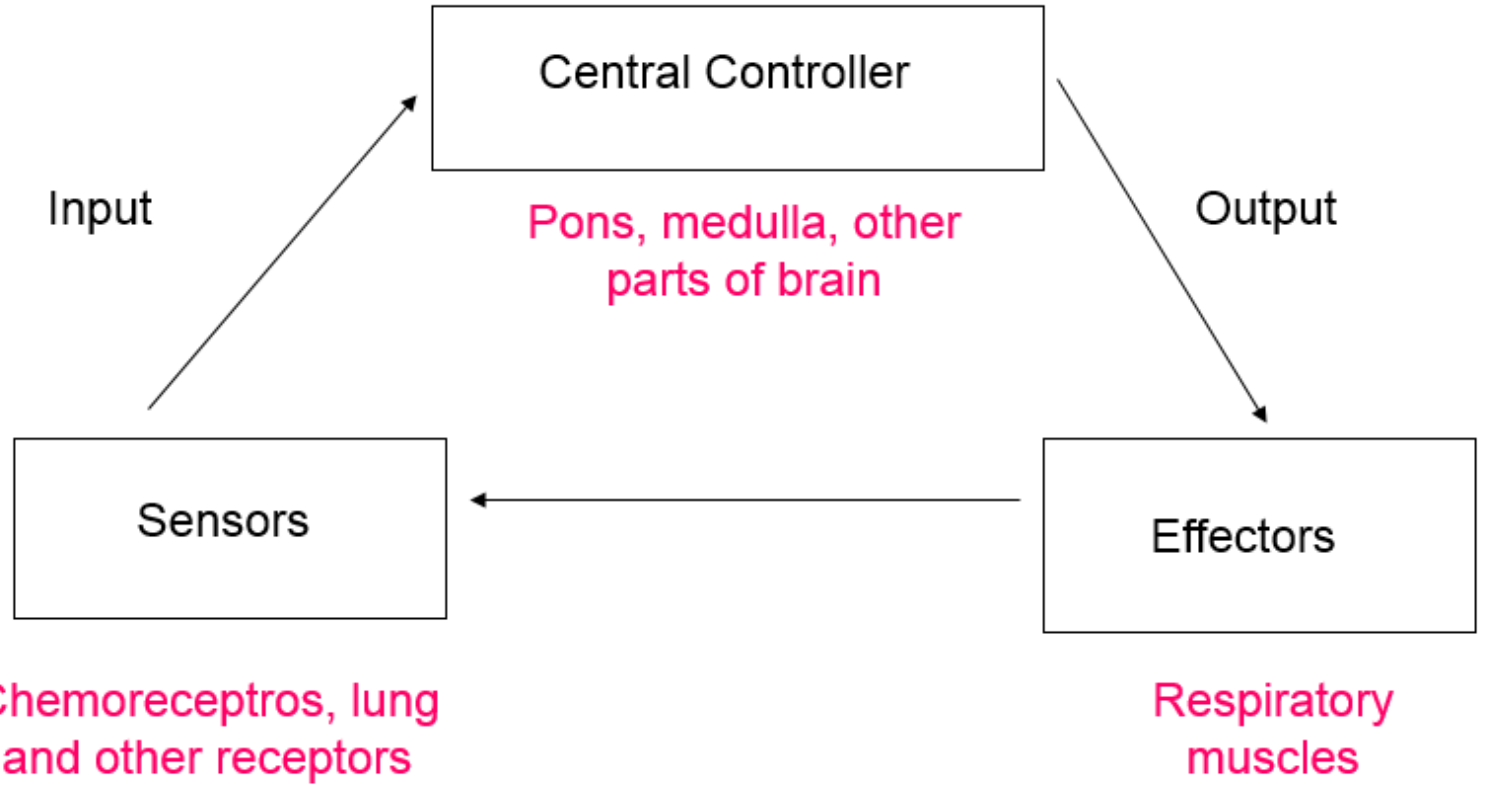
The Aim of Ventilation

- To provide adequate O_2 to the tissues dependant on need
- To provide effective excretion of CO_2
- To maintain pH homeostasis of the blood plasma
- To facilitate flexible functionality





Respiratory Control:



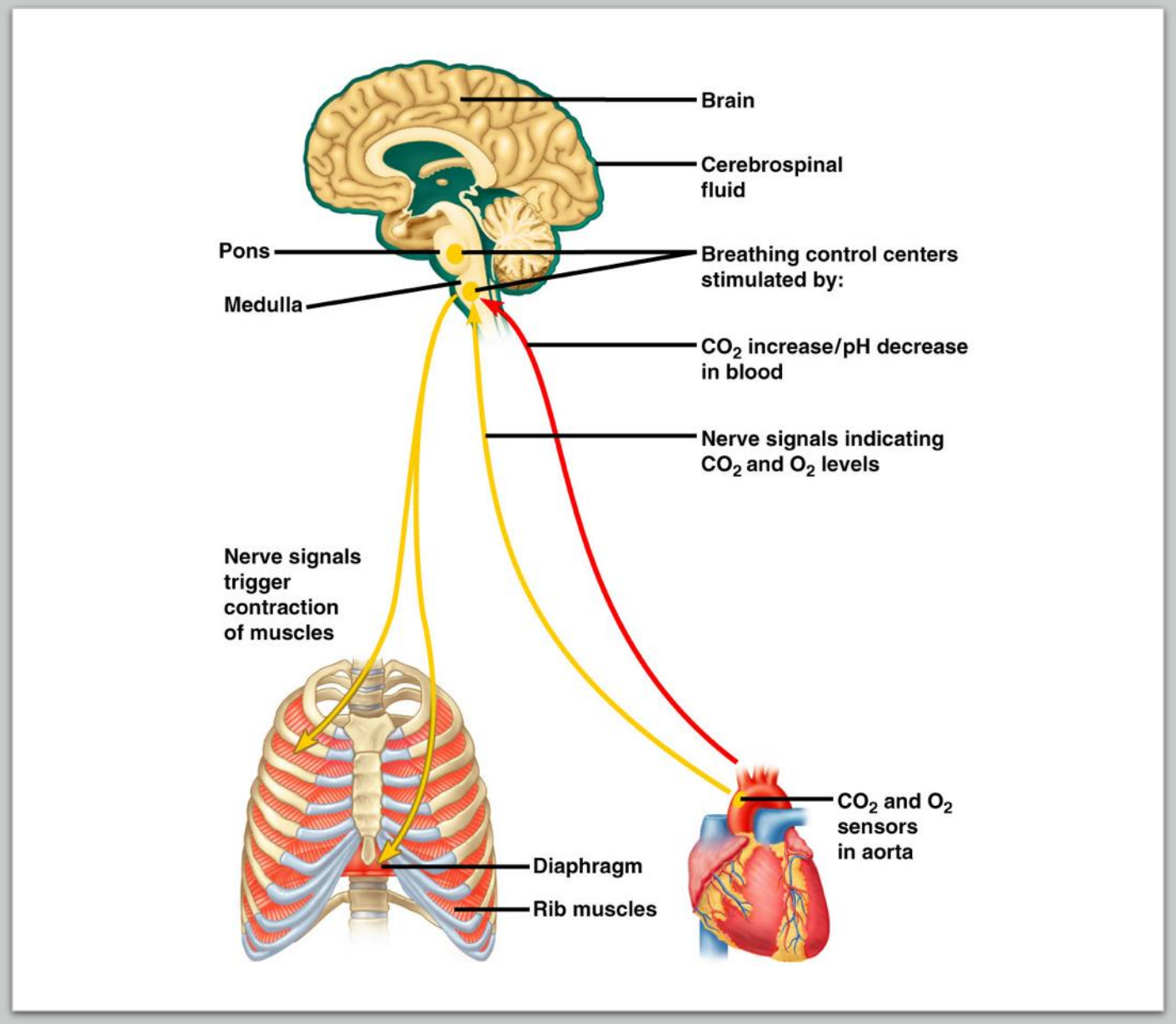
Regulation of Respiration:

Mechanisms of regulation of respiration:

- Nervous regulation
- Chemical regulation

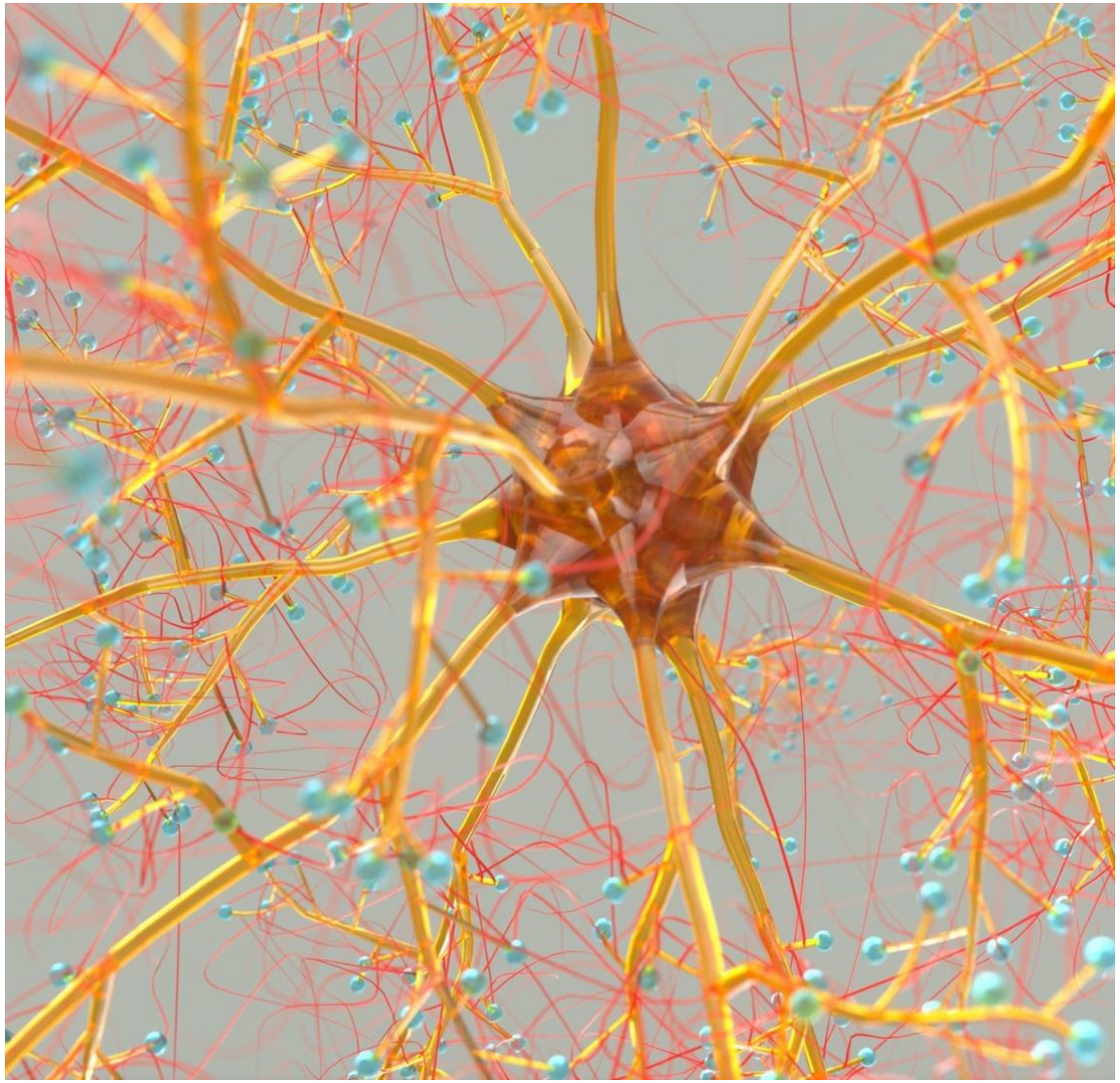
Purpose of respiration regulation:

- Maintain constant level of O2 and CO2
- Adjust O2 supply as per metabolic demand of body
- Regulate acid base balance (pH)



Sensors:

- Central chemoreceptors
- Peripheral chemoreceptors (most important)
- Lung receptors
- Other receptors



Central vs Peripheral Chemoreceptors:

	Central Chemoreceptors	**Peripheral chemoreceptors**
Location	Ventral surface of brainstem	Aortic arch and carotid body
What are they sensitive to	Directly: pH of CSF Indirectly: CO_2 in plasma	Directly: O_2, CO_2, and H^+ (pH)
Influence	"respiratory drive" or drive to breathe	Acute changes in blood gases
Effects	↑ respiratory rate/ depth in response to hypercapnia (slower than peripheral)	↑ respiratory rate/ depth in response to hypercapnia, hypoxia, or acidosis

Chemical regulation of respiration:

3 CHEMICAL FACTORS:

CONCENTRATION OF CO2 IN THE BLOOD

CONCENTRATION OF H+ IONS IN THE BLOOD OR pH

CONCENTRATION OF O2 IN THE BLOOD

CO2 stimulates chemoreceptors
2 groups – peripheral & central

When CO_2 concentration in blood increases

⇩

Stimulates the chemoreceptors

⇩

Transmission of sensory impulses to respiratory centers

⇩

Activation of respiratory centers

⇩

Increases the activities of respiration (rate and Depth)

⇩

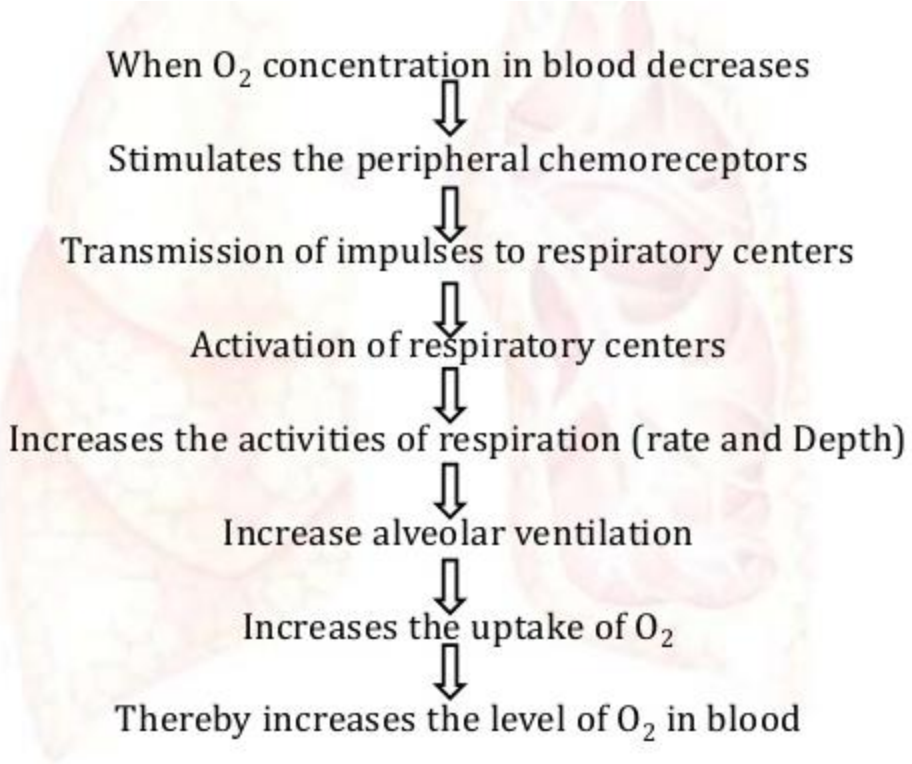
Increase alveolar ventilation

⇩

Expulsion of CO_2 and decreases the level of CO_2 in blood

Chemical regulation of respiration:

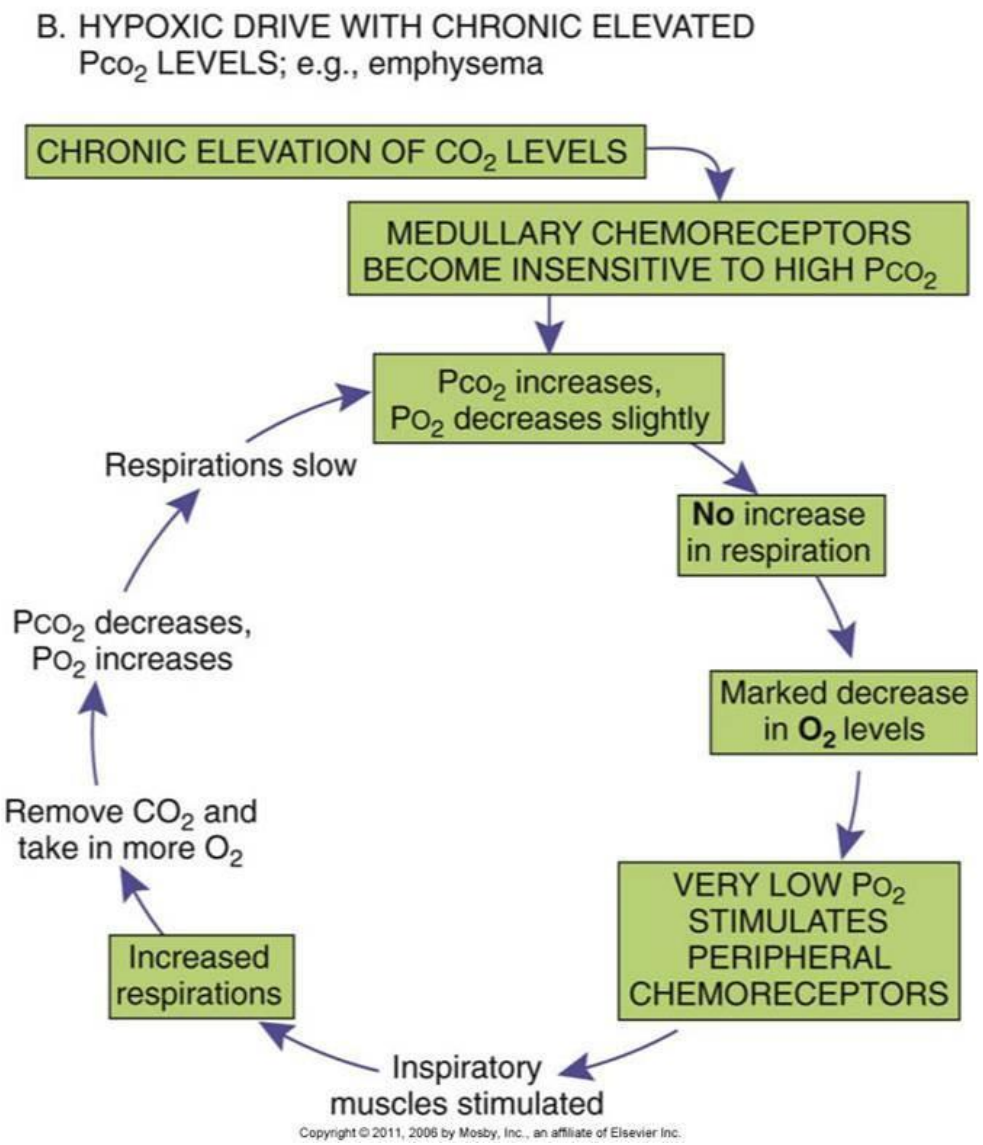
CONCENTRATION OF O2 IN THE BLOOD



CONCENTRATION OF H+ IONS IN THE BLOOD OR pH

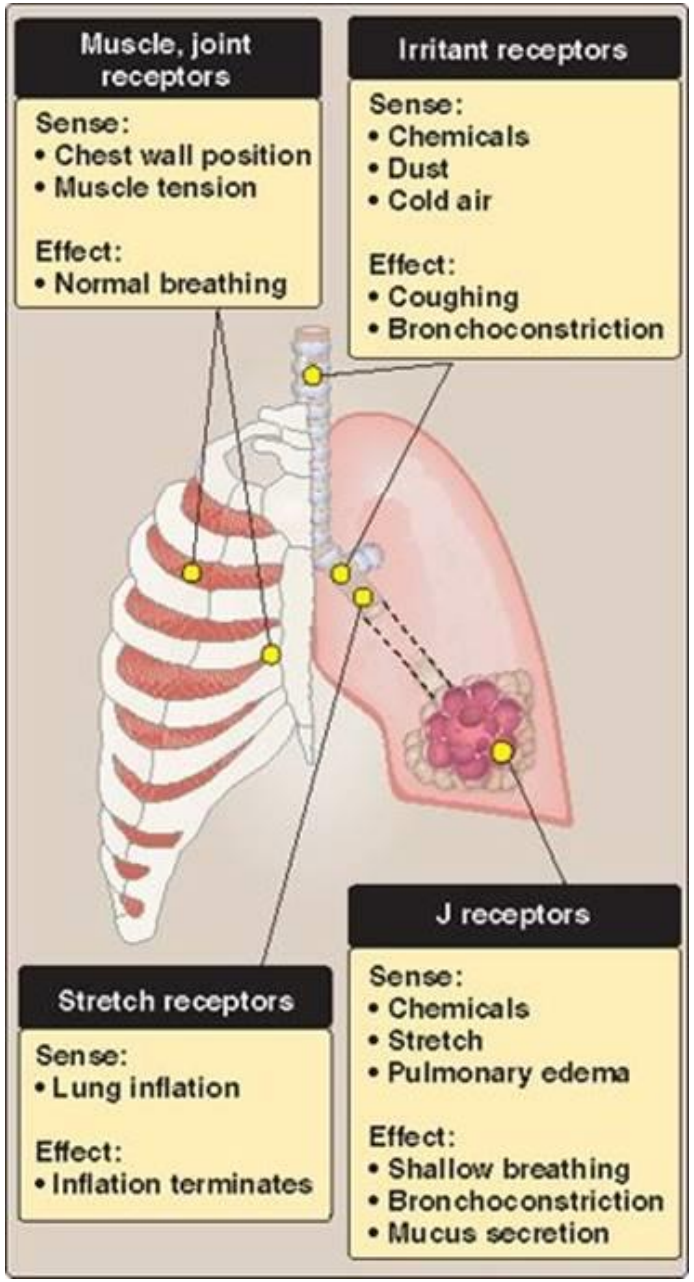
- Peripheral chemoreceptors normally involved in response to pH
- If pH is very low, can seep through blood brain barrier and affect central chemoreceptor
- Difficult to distinguish from action of increased pCo2
- Low pH = increased ventilation

Hypoxic Drive



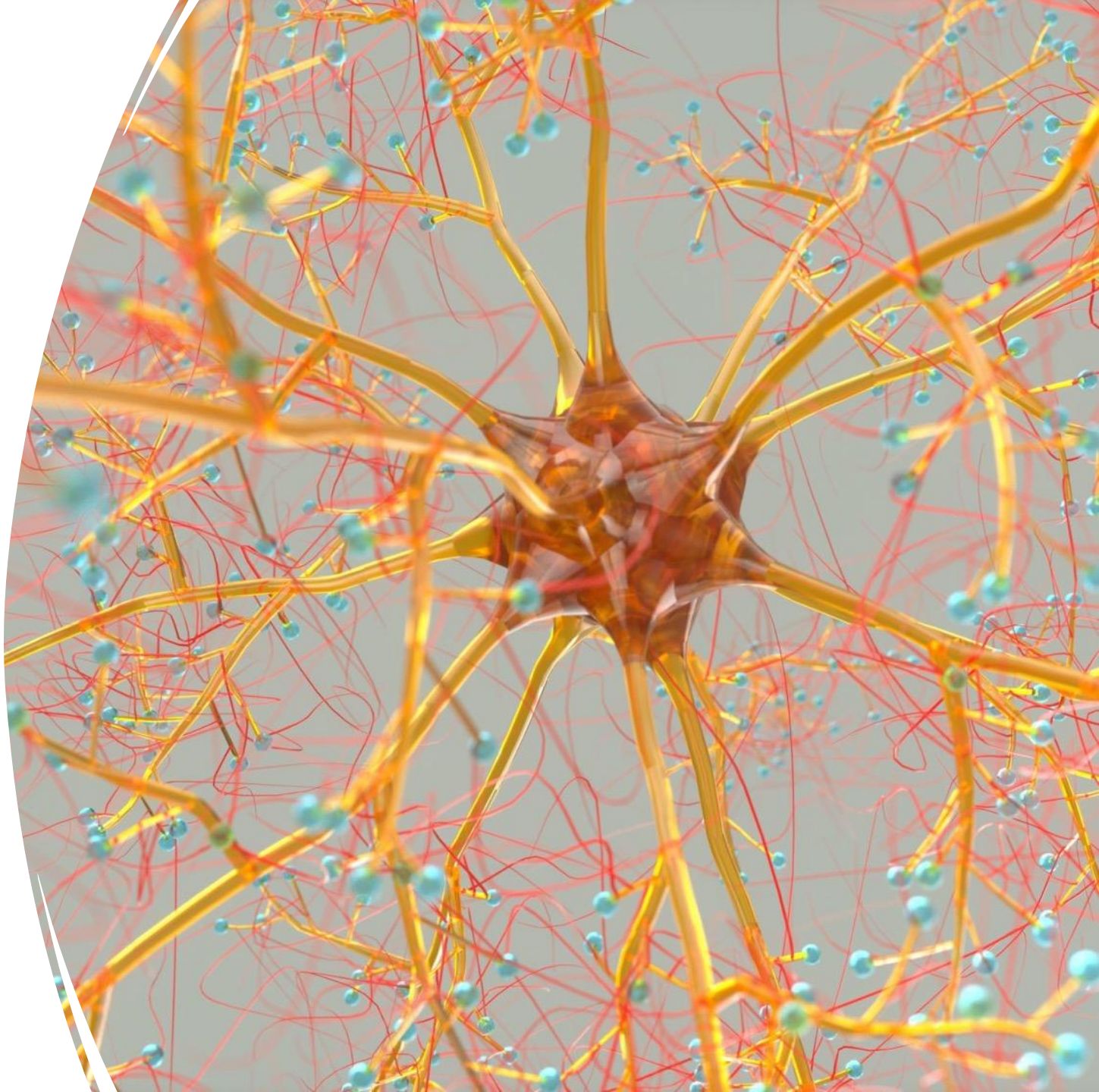
Lung Receptors:

- Pulmonary Stretch Receptors
- Irritant Receptors
- J Receptors
- Bronchial C fibres



Other receptors:

- Nose and upper airway receptors
- Joint and muscle
- Gamma system
- Pain and temp
- Arterial baroreceptors



Summary

- Many factors affect respiratory centre activity to ensure:
  - Adequate ventilation
  - ABG profile maintained within an acceptable range
- Every cell must have O_2
- Every cell produces CO_2 to be excreted
- CO_2 is the most profound stimulus to breath in health
- CO_2 works primarily on central chemoreceptors via H^+ and secondly via peripheral chemoreceptors
- In health, pO_2 only influences ventilation via enhancing sensitivity of the central chemoreceptors to pCO_2
- In disease, a pO_2 of < 8kPa becomes the prime stimulus to breathe

RESPIRATORY TERMINOLOGIES:

HYPOXIA	Decrease in O2 supply to tissues & cells
HYPOXAEMIA	Decrease in arterial oxygen tension in the blood
HYPERCAPNIA	Increase in partial pressure of CO2 in the blood
APNEA	Temporary cessation of breathing
TACHYPNEA	Abnormally rapid breathing
BRADYPNEA	Abnormally slow breathing
DYSPNEA	Laboured breathing / shortness of breath

FURTHER RESPIRATORY TERMINOLOGIES:

- **Apnea:** temporary cessation of breathing
- **Tachypnea:** abnormally rapid respirations
- **Bradypnea**: abnormally slow respiration
- **Dyspnea:** labored breathing or shortness of breath
- **Hypoxemia:** decrease in arterial oxygen tension in the blood
- **Hypoxia:** decrease in oxygen supply to the tissues and cells
- **Hypercapnia:** an increase in the partial pressure of carbon dioxide in the blood.
- **Hypocapnia:** a decreased amount of carbon dioxide in the blood.
- **Physiologic dead space:** portion of the tracheobronchial tree that does not participate in gas exchange.
- **Central cyanosis:** bluish discoloration of the skin or mucous membranes due to hemoglobin carrying reduced amounts of oxygen.
- **Intrapleural (intrathoracic) pressure:** pressure between the two pleural layers in the pleural cavity.
- **Alveolar (intrapulmonic) pressure:** The pressure within the tiny gas-exchanging structures of the lungs.
- **Diffusion:** exchange of gas molecules from areas of high concentration to areas of low concentration.
- **Osmosis:** The passage of solvent through a semipermeable membrane that separates solutions of different concentrations.
- **Dalton's law:** (John Dalton, Brit. chemist, 1766–1844) A law that states that, in a mixture of gases, the total pressure is equal to the sum of the partial pressures of each gas.
- **Henry's law: (**William Henry, Brit. chemist, 1774–1836) The weight of a gas dissolved by a given volume of liquid at a constant temperature is directly proportional to the pressure.
- **Boyle's law: (**Robert Boyle, Brit. physicist, 1627–1691) A law stating that, at a constant temperature, the volume of a gas varies inversely with the pressure.

Chronic Obstructive Pulmonary Disease (COPD):

The **GOLD initiative** defines COPD as :

"A heterogenous lung condition characterized by chronic respiratory symptoms (dyspnoea, cough, sputum production) due to abnormalities of the airways (bronchitis, bronchiolitis) &/or alveoli (emphysema) Which cause persistent, often progressive, airflow obstruction."

A disease state characterised by airflow obstruction resulting from chronic bronchitis or emphysema

Airway obstruction is progressive

Airway obstruction is only partially reversible and does not change markedly over several months

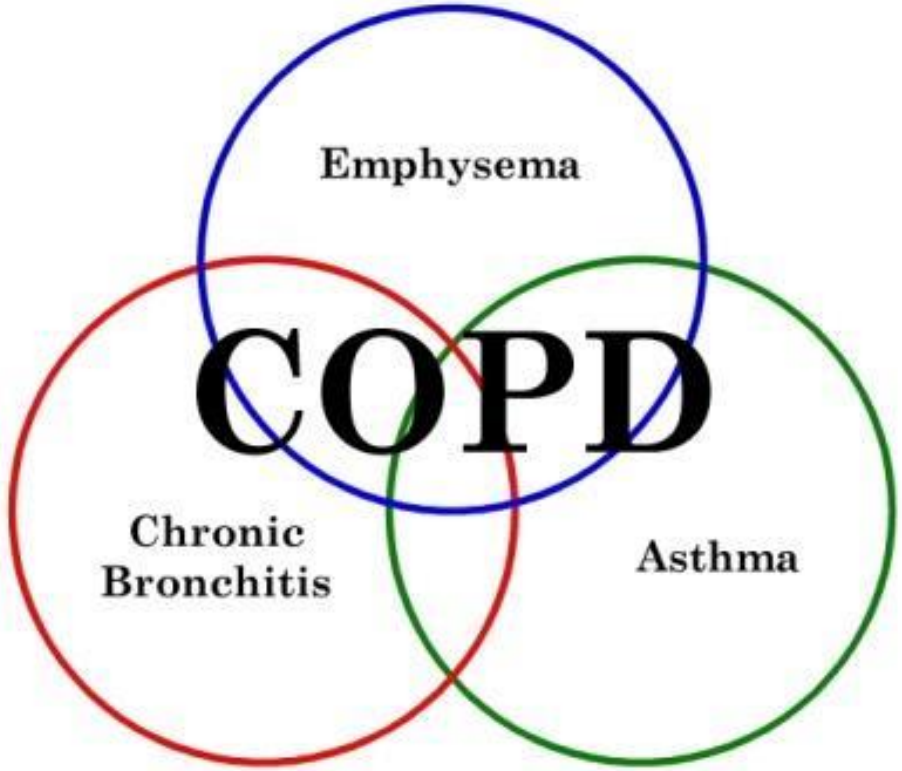


Table 4. Common Causes of Obstructive and Restrictive Lung Disease

Obstructive

- α_1-antitrypsin deficiency
- Asthma
- Bronchiectasis
- Bronchiolitis obliterans
- Chronic obstructive pulmonary disease
- Cystic fibrosis
- Silicosis (early)

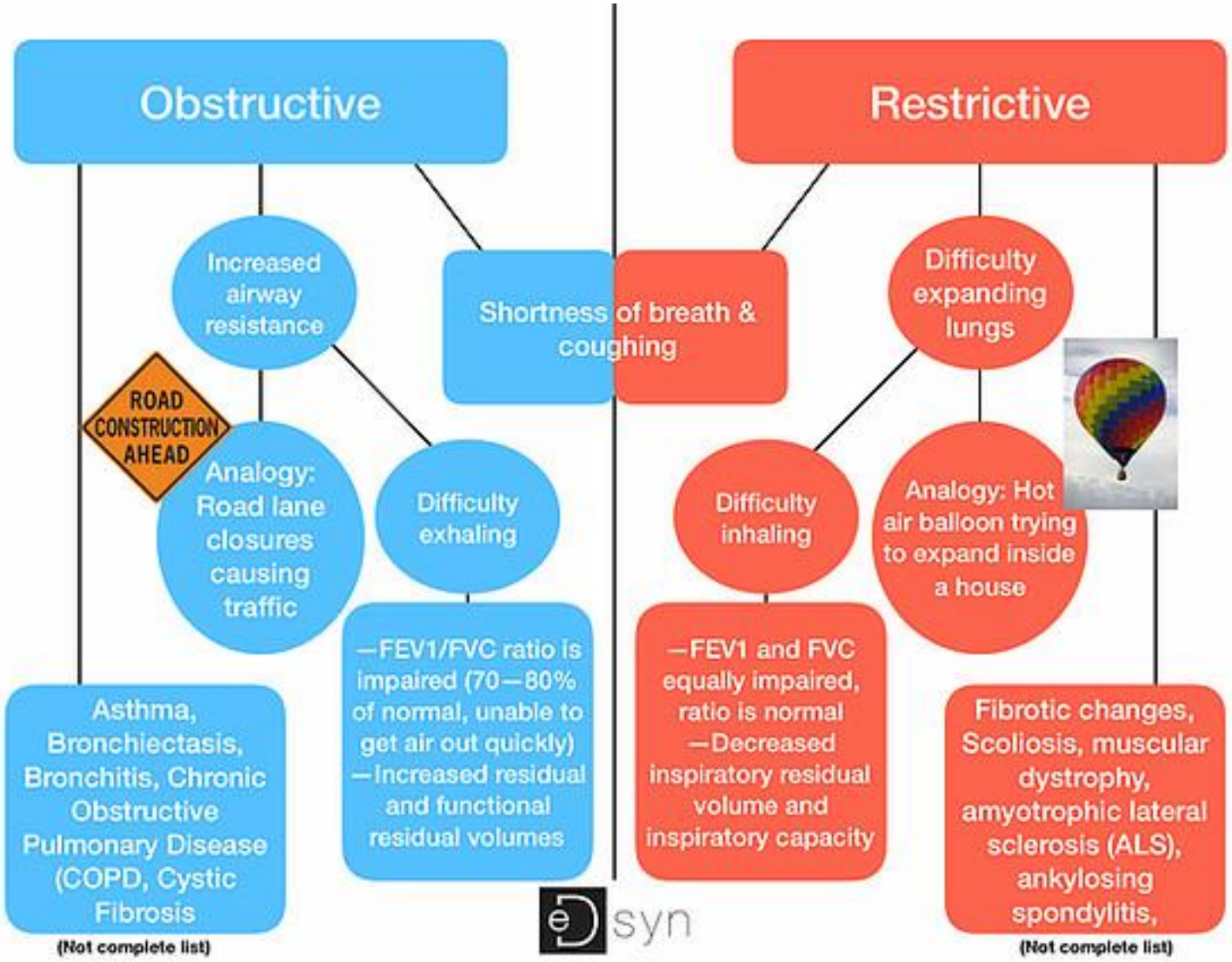
Restrictive

- Chest wall
  - Ankylosing spondylitis
  - Kyphosis
  - Morbid obesity
  - Scoliosis
- Drugs (adverse reaction)
  - Amiodarone
  - Methotrexate
  - Nitrofurantoin (Furadantin)

Restrictive *(continued)*

- Interstitial lung disease
  - Asbestosis
  - Berylliosis
  - Eosinophilic pneumonia
  - Hypersensitivity pneumonitis
  - Idiopathic pulmonary fibrosis
  - Sarcoidosis
  - Silicosis (late)
- Neuromuscular disorders
  - Amyotrophic lateral sclerosis
  - Guillain-Barré syndrome
  - Muscular dystrophy
  - Myasthenia gravis

Information from references 20 through 35.



Epidemiology of COPD:

COPD is laden with gloomy statistics :

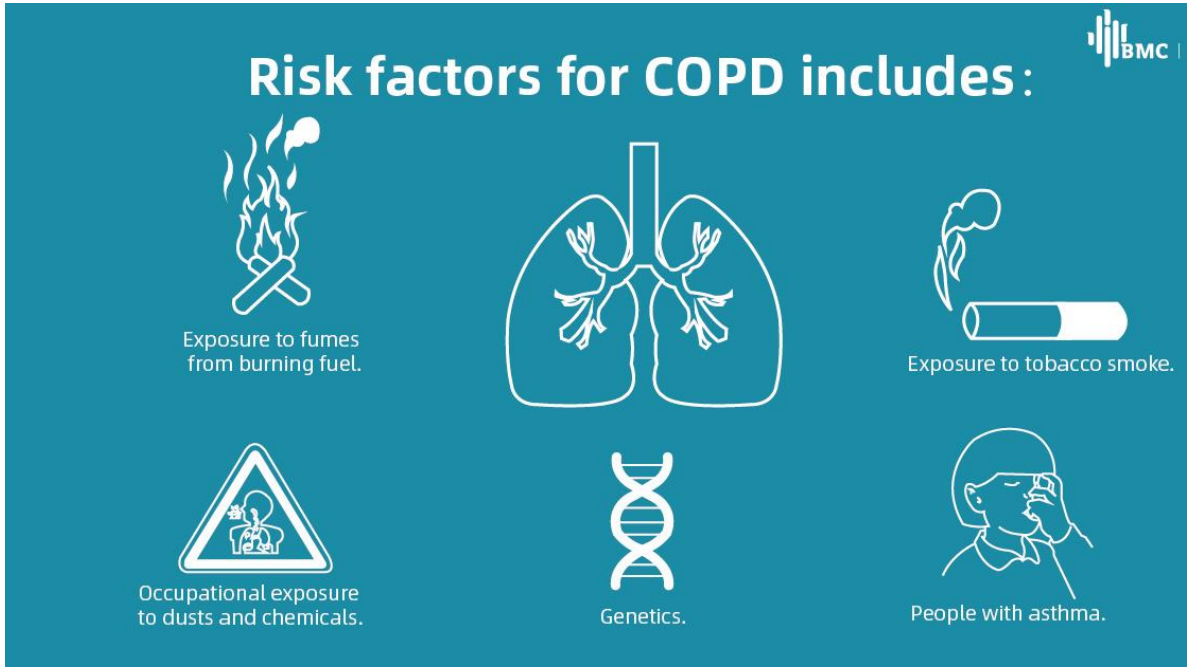
- Globally around 3 million deaths annually due to COPD
- Expected to increase to 5.4 million by 2060

- Asthma + Lung UK (previously known as BLF UK) estimated 1.2 million people (2% of the population) are living with COPD (2019)

- Affects roughly 10% of the global population (Mulhall 2016)

- Half of elderly smokers have the disease (Lindberg et al 2016)

- Huge financial burden on the NHS
- Respiratory disease accounts for approximately 6% of the total healthcare budget



Causes of COPD

- Usually caused by smoking (Passive smoking and in utero exposure to smoking can cause it too)
- Air Pollution
- Inhaling dust or fumes at work (occupational exposure)
- Respiratory infections
- Low Birthweight , Malnutrition and Childhood Respiratory Illness (Soto-Martinez 2010)

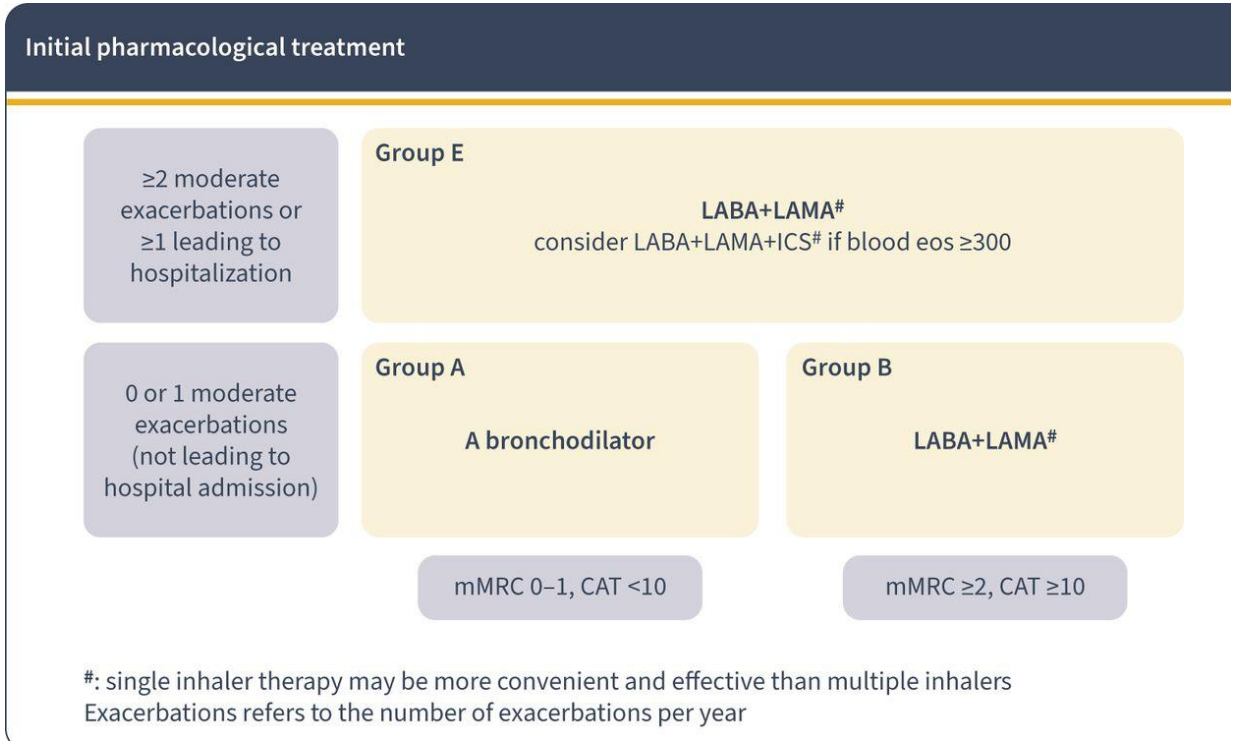
Diagnosing COPD

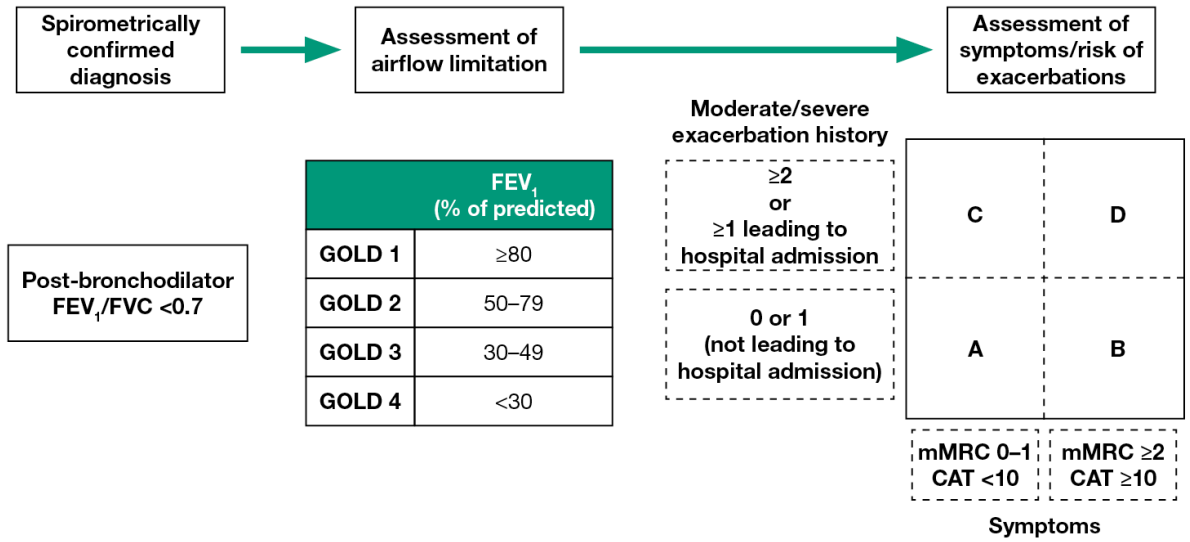
- Subjective history exploring smoking habits, discussing frequency of chest infections , symptoms (particularly shortness of breath and chronic productive cough)
- Spirometry readings which help diagnose and stratify the severity of COPD.
- Relevant imaging e.g. Chest XRAYs or CT Chest to establish key clinical features





CATEGORISATION OF COPD



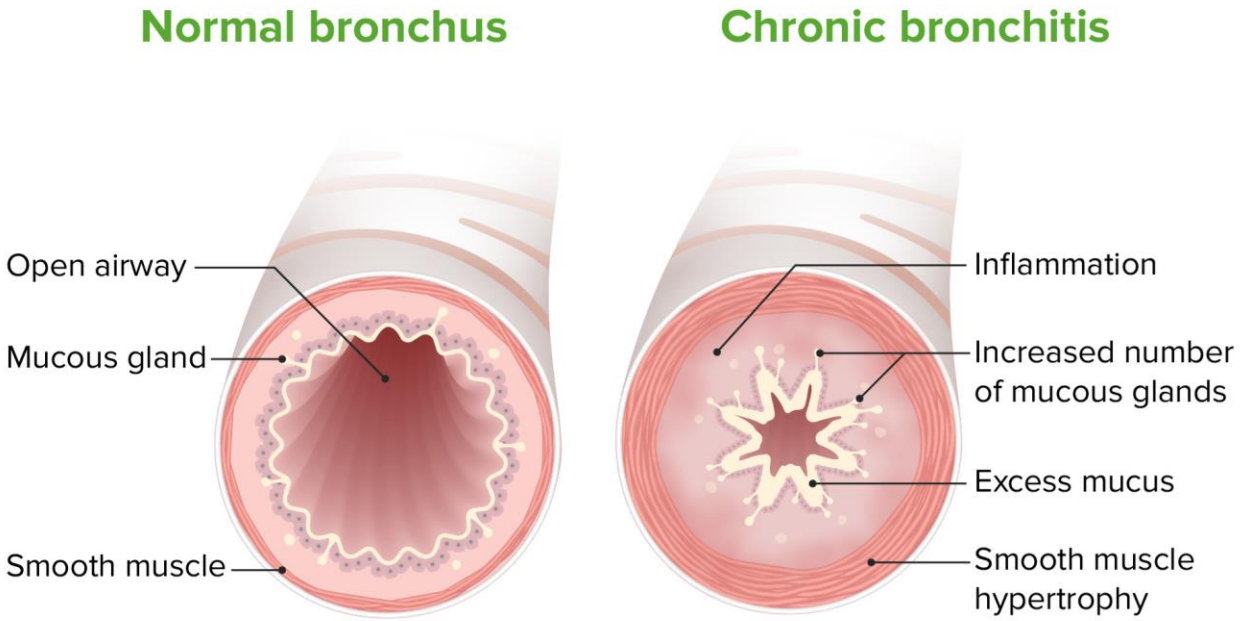


FEV_1=forced expiratory volume in the first second; FVC=forced vital capacity; mMRC=modified Medical Research Council; CAT=COPD assessment test.

Chronic Bronchitis:

- Chronic bronchitis can be defined **as a type of COPD** whereby recurrent inflammation of the bronchioles occur due to a productive cough lasting *more than 3 months occurring within a span of 2 years*.

- Other causes of symptoms, such as tuberculosis or other lung diseases, must be ruled out.



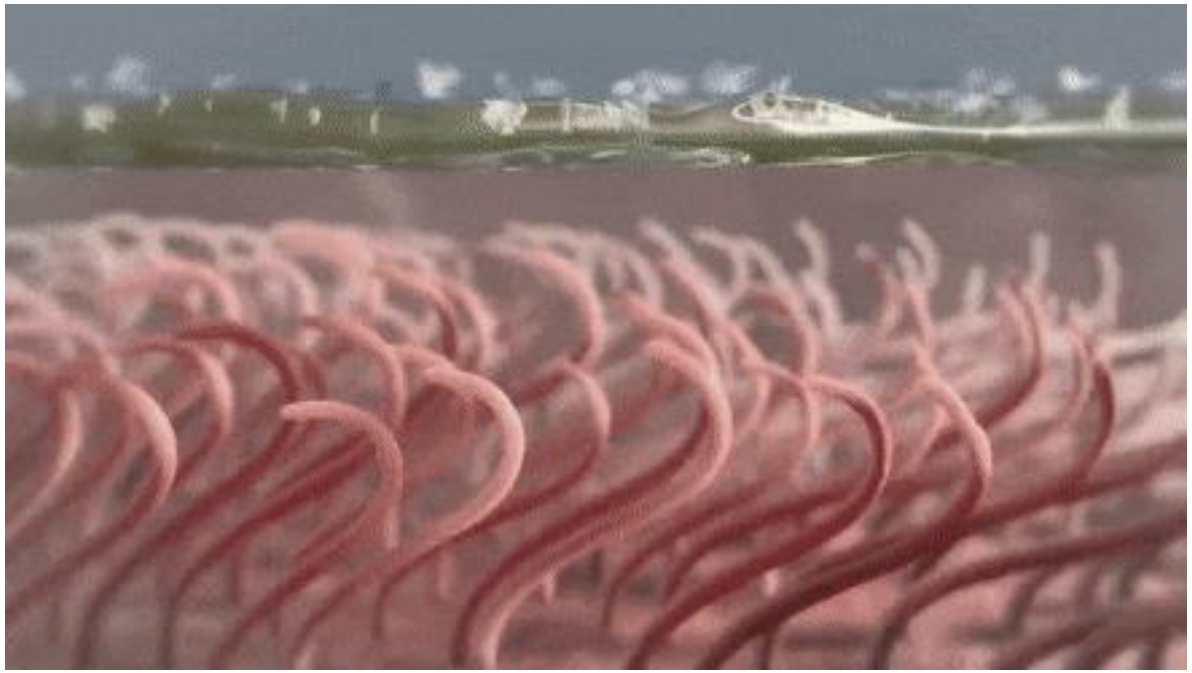
Causes of Chronic Bronchitis

- **Cigarette smoke.** People who smoke or who live with a smoker are at higher risk of both acute bronchitis and chronic bronchitis.
- **Low immune resistance.** This may result from another acute illness, such as a cold, or from a chronic condition that compromises your immune system. Older adults, infants and young children have greater vulnerability to infection.
- **Exposure to irritants on the job.** Your risk of developing bronchitis is greater if you work around certain lung irritants, such as grains or textiles, or are exposed to chemical fumes.
- **Gastric reflux.** Repeated bouts of severe heartburn can irritate your throat and make you more prone to developing bronchitis.





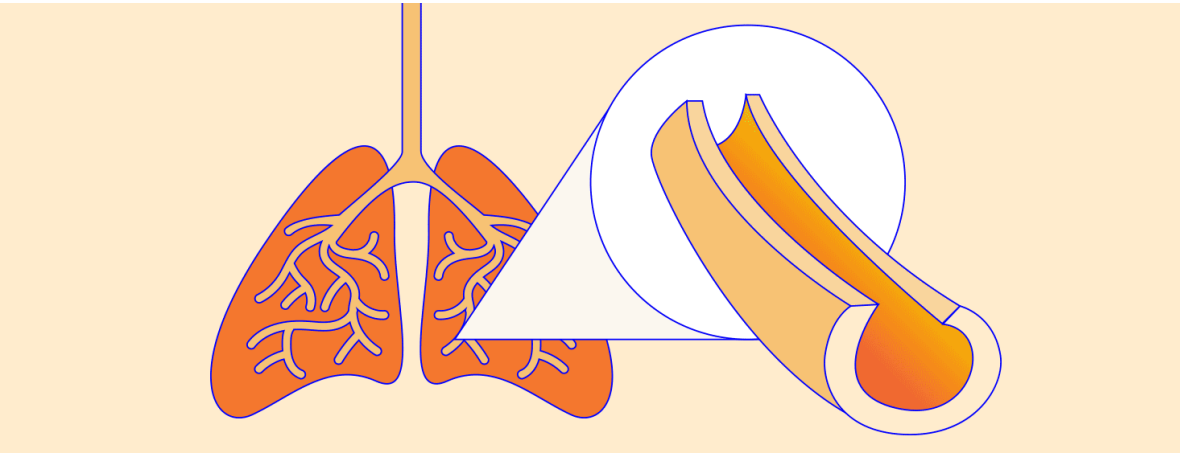




Chronic Bronchitis

- Repeated inhalation of pollutants leads to irritation of the airway mucosa
- Hyperplasia and hypertrophy of mucus glands in the large airways
- Excessive amounts of mucus in the airways
- Hyperplasia of goblet cells occurs at the expense epithelial cells
- Reduced ciliary function
- Small airways become obstructed by mucus
- Bacterial colonisation and repeated infection

Chronic Bronchitis



- Small airways become narrowed due to inflammatory changes – airway remodelling
- Mucosal oedema
- Smooth muscle hypertrophy
- Peribronchial fibrosis

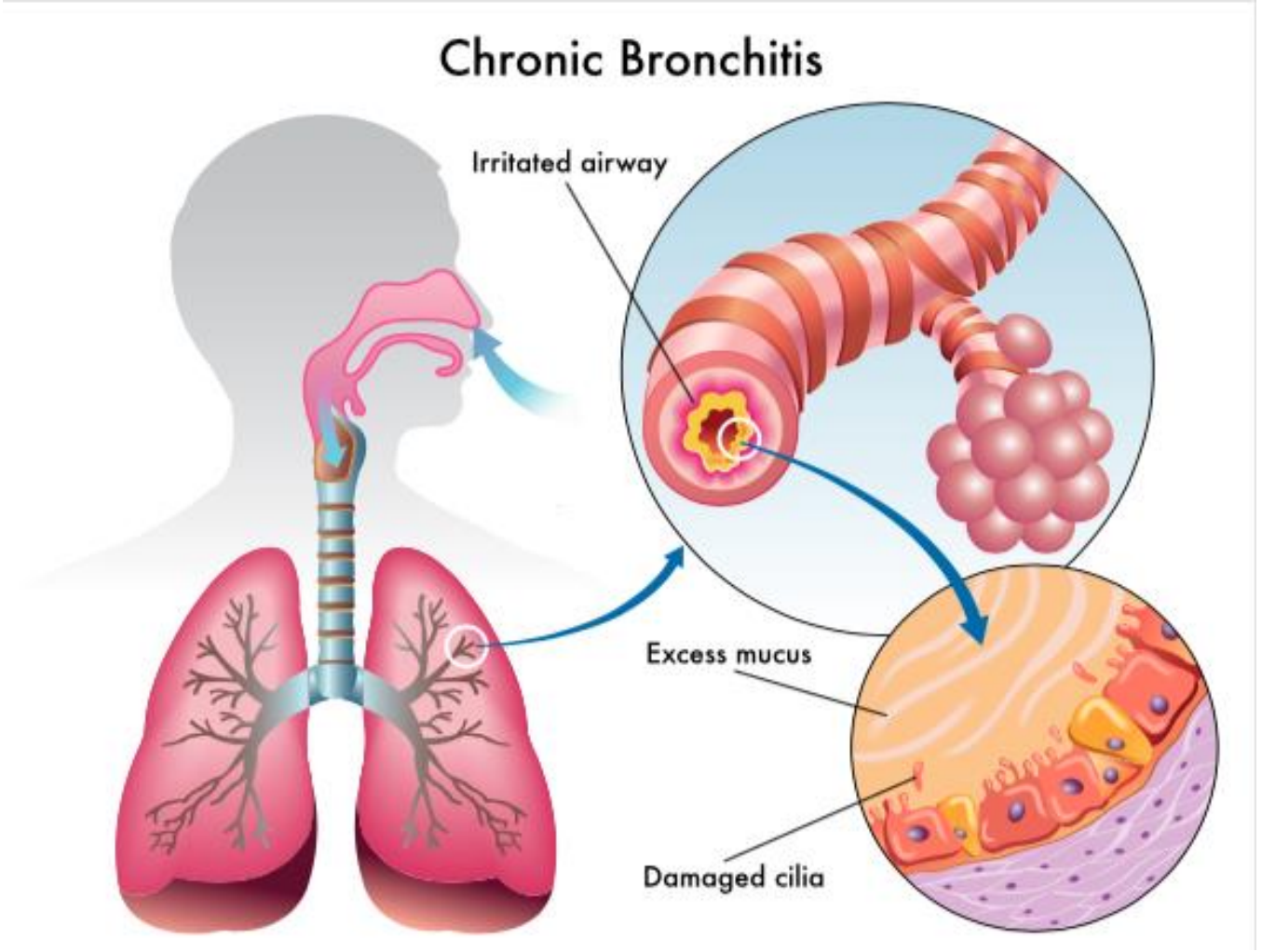
Pathophysiology and Clinical Features

Key Clinical Features:

- Airway obstruction & resistance to airflow
- Excess mucus secretions
- Impaired gas exchange and V/Q mismatch
- Productive cough
- Wheeziness / Chest tightness
- Recurrent infections / exacerbations
- Increased work of breathing

Secondary Features:

- Urinary Incontinence
- Malaise/ Fatigue



Diagnosing Chronic Bronchitis

Subjective History: complete history from the patient including information regarding possible exposure to inhaled irritants or chemicals as well as full details regarding smoking habits.

Fever is uncommon in chronic bronchitis and when present can be suggestive of associated influenza or pneumonia.

- Exploring timeline of symptoms is key to ascertain if individual is suffering from Chronic Bronchitis

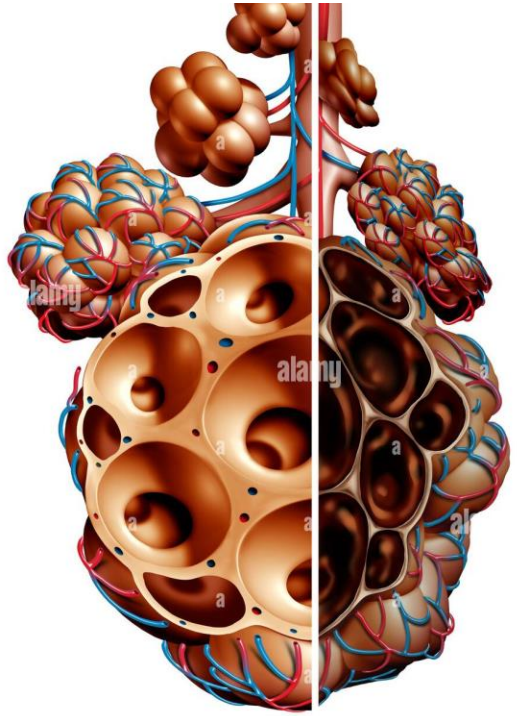
Spirometry

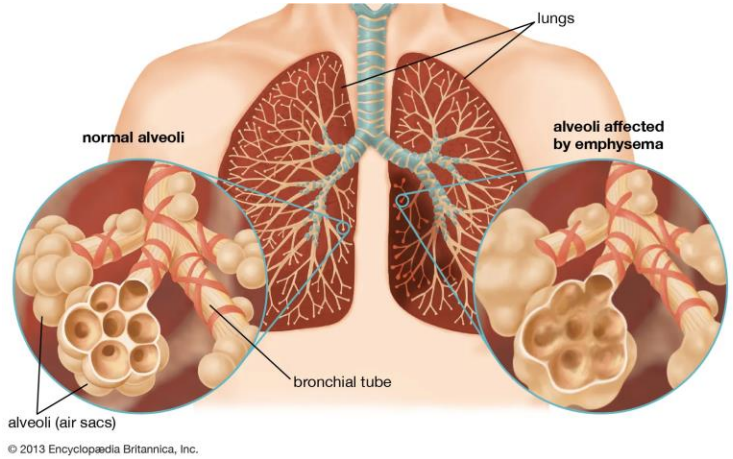
CXR/ CT Chest

Emphysema:

- Emphysema is characterized by abnormal permanent enlargement of alveoli with the destruction of their walls (without any fibrosis) and destruction of lung parenchyma with loss of elasticity.

- ***<u>There are secondary effects on larger airways.</u>***





Aetiology of Emphysema

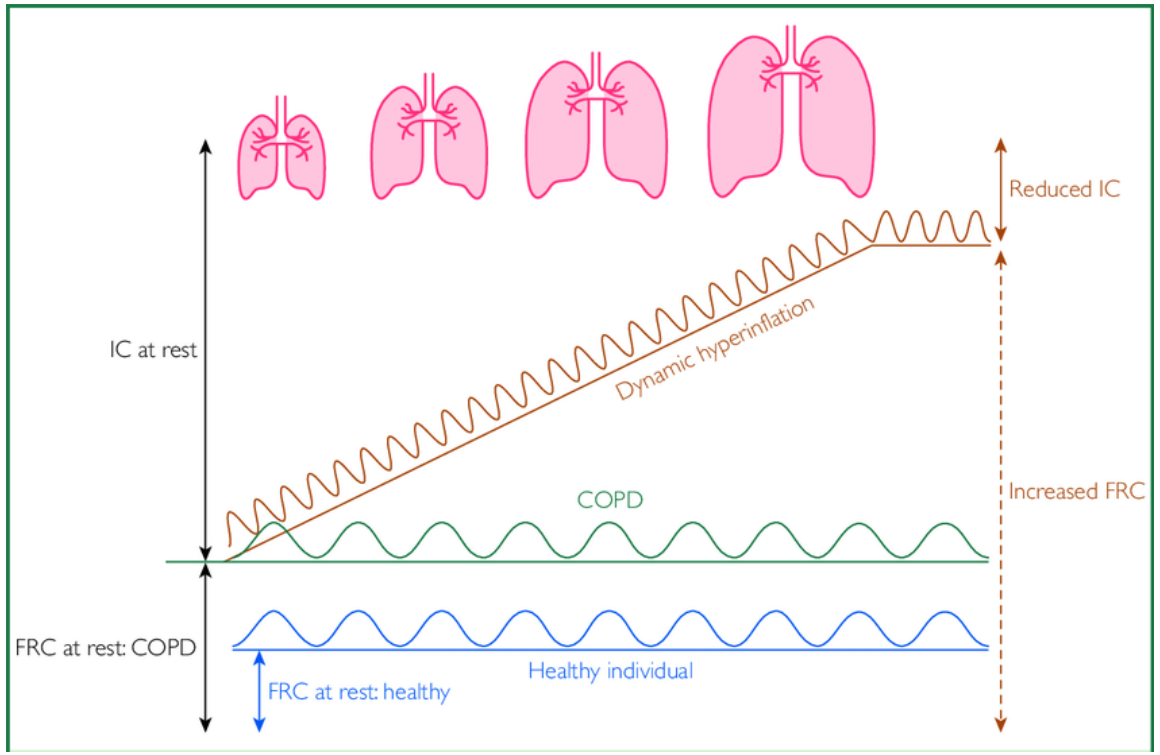
- **SMOKING is the major cause**

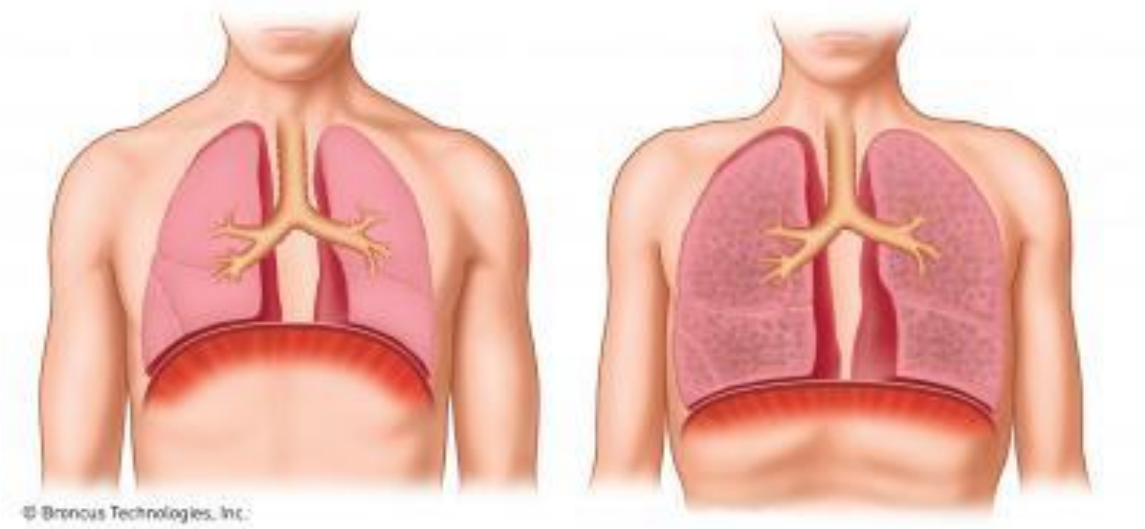
- Alpha 1 antitrypsin deficiency
  - Genetic cause
  - Alpha 1 anti-trypsin inhibits the action of neutrophil elastase which destroys elastin
  - Develops before the age of 40
  - Equal distribution between sexes



Pathophysiology

- Proteolytic degradation of the alveolar walls and destruction of elastic fibres
- In the alveoli, erosion of alveolar septum breaking down into bullae
- Distally, bronchioles are destabilizing by loss of the elastic recoil which normally splints them open. Therefore, airways become floppy, can collapse and cause gas trapping.
- Hyperinflation





Secondary effects of emphysematous changes

- Airflow obstruction
- Increased resistance to airflow
- Reduced surface area for gas exchange
- V/Q mismatch
- Increased work of breathing



CHRONIC BRONCHITIS & EMPHYSEMA MAY CO-EXIST



Clinical Features of COPD

- Chronic Cough
- Accessory Muscle Overactivity
- Shortness of breath (SOB) on exertion
- Reduced exercise tolerance
- Hypoxia/ Hypercapnia in severe cases
- Hyperinflation
- Weight loss
- Cyanosis
- Clubbing

CHRONIC BRONCHITIS

CLINICAL DIAGNOSIS: DAILY PRODUCTIVE COUGH FOR THREE MONTHS OR MORE, IN AT LEAST TWO CONSECUTIVE YEARS

OVERWEIGHT AND CYANOTIC

ELEVATED HEMOGLOBIN



PERIPHERAL EDEMA

RHONCHI AND WHEEZING

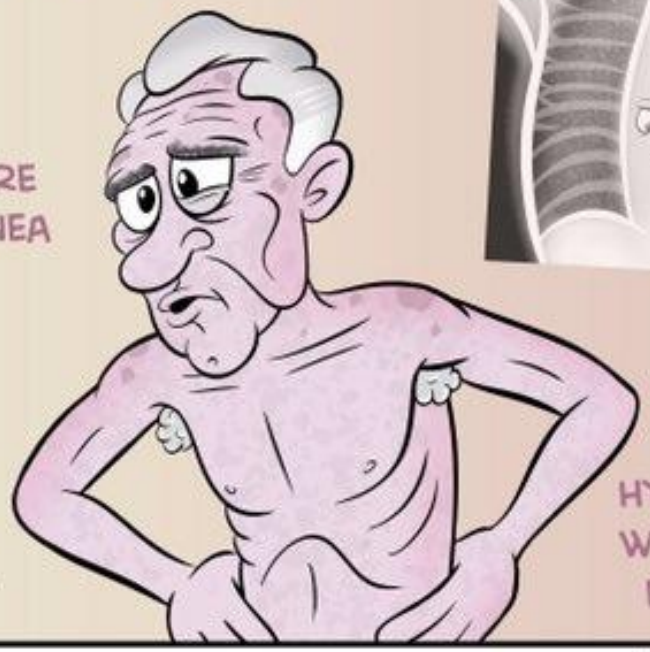
EMPHYSEMA

PATHOLOGIC DIAGNOSIS: PERMANENT ENLARGEMENT AND DESTRUCTION OF AIRSPACES DISTAL TO THE TERMINAL BRONCHIOLE

OLDER AND THIN



SEVERE DYSPNEA



QUIET CHEST

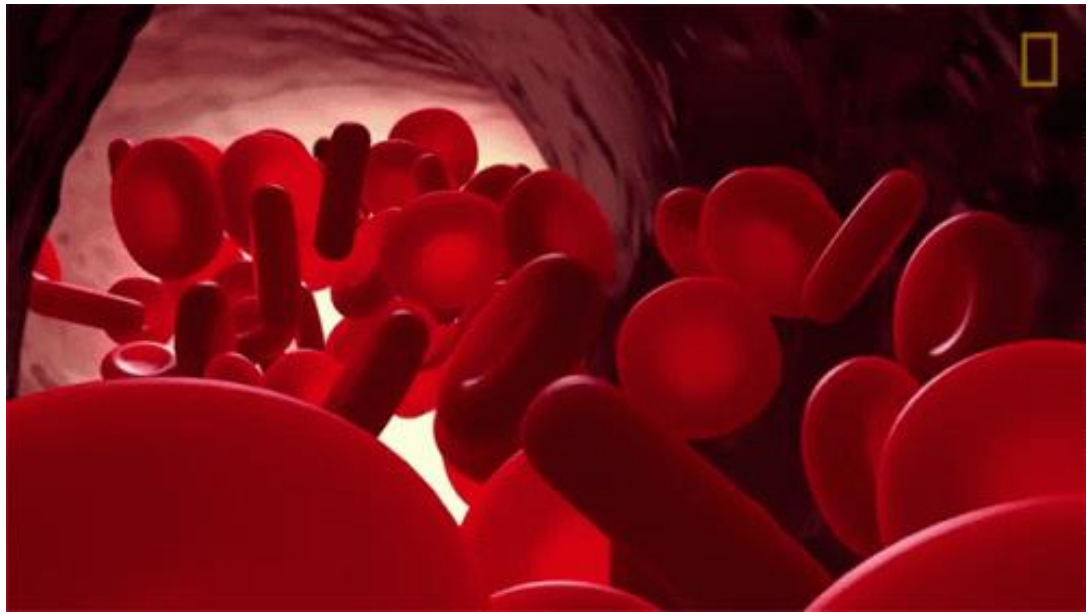
X-RAY: HYPERINFLATION WITH FLATTENED DIAPHRAGMS

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Further secondary effects of COPD

- Cor Pulmonale
  - Right ventricular failure

- Polycythaemia
  - Bodies response to chronic hypoxia
  - Increase in total volume of red blood cells
  - Increased blood viscosity



Chest X-ray:

- Emphysema
- Hyperinflated
- Long thin heart
- Black lung fields

- Chronic Bronchitis
- Patchy shadowing
- Increased cardiac shadow







Signs of Respiratory Distress in COPD

- Pursed lip breathing
- Prolonged expiration
- Increased RR
- Cyanosis
- Active expiration
- Fixing of upper limbs
- Use and hypertrophy of accessory muscles
- Soft tissue recession
- Hoover's sign

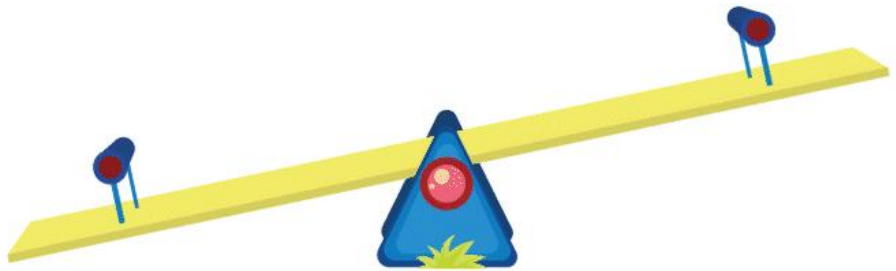


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Respiratory Failure

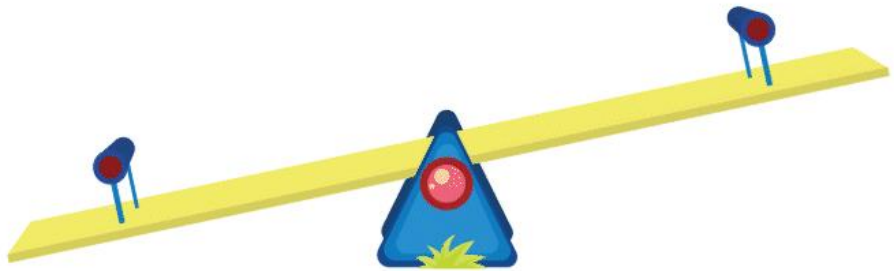
- The inability to maintain partial pressures in arterial blood of carbon dioxide ($PaCO_2$) and oxygen (PaO_2) within normal physiological limits

- $PaO_2 < 8kPa$
- $PaCO_2 > 6.5$ kPa



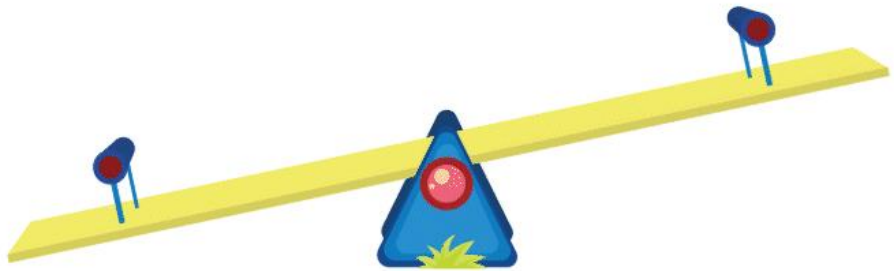
Type 1 Respiratory Failure

- Decreased PaO_2
- Normal or low $PaCO_2$
- Treated with supplementary oxygen

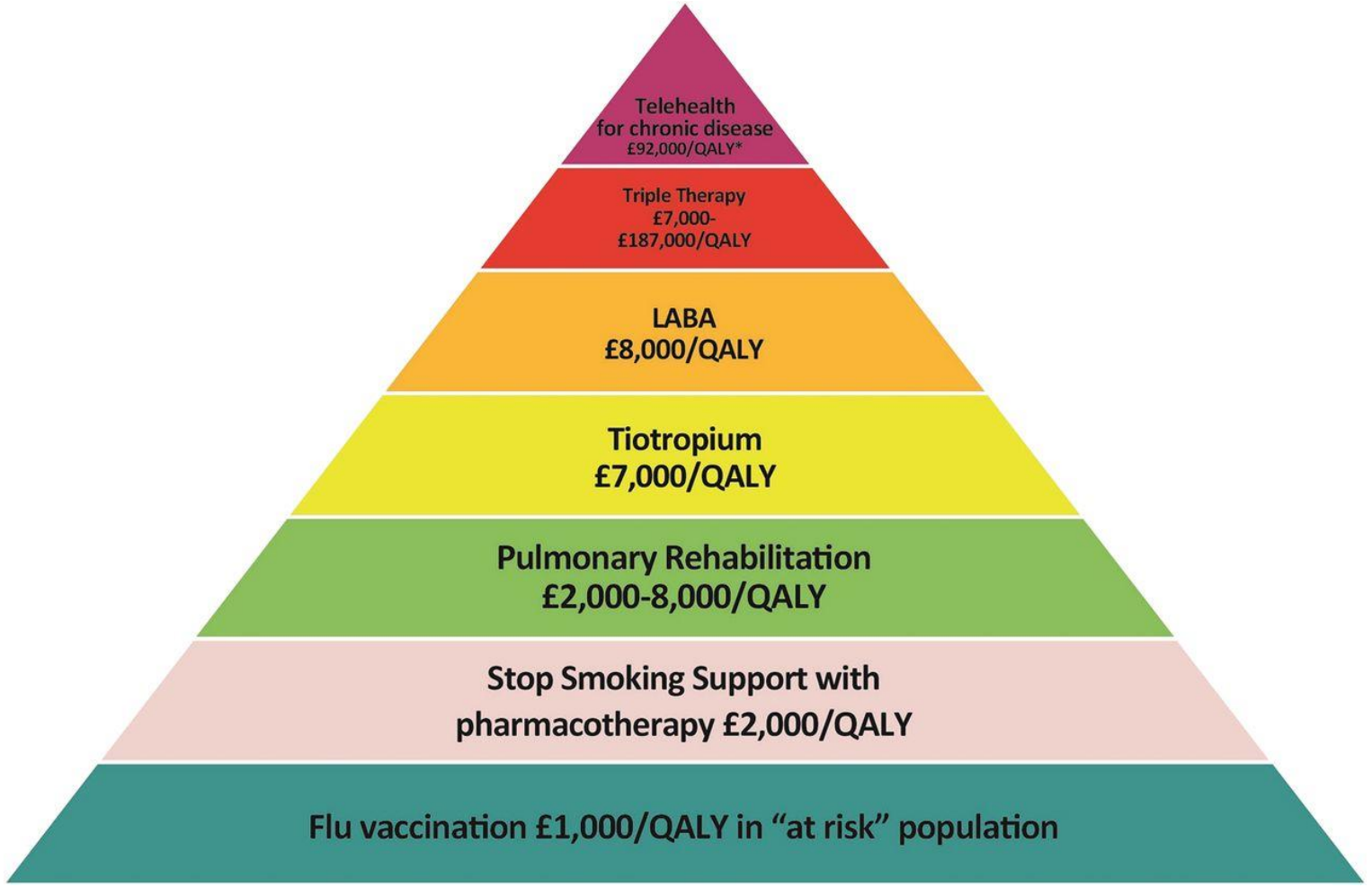


Type 2 Respiratory Failure

- Decreased PaO_2
- Increased $PaCO_2$
- Alveola hypoventilation
- Can be:
  - Acute
  - Chronic
  - Acute on Chronic



Management of stable COPD



Further management:

Long term oxygen therapy

- Assessment ideally carried out in a stable state
- 2 assessments required minimum of 3 weeks apart (if hypoxic during first appointment)
- To use for a minimum of 15HPD

Non-invasive ventilation

- T2RF

Surgery

- Endobronchial valves
- Coils
- Thermal ablation
- Lung volume reduction surgery
- Lung transplant

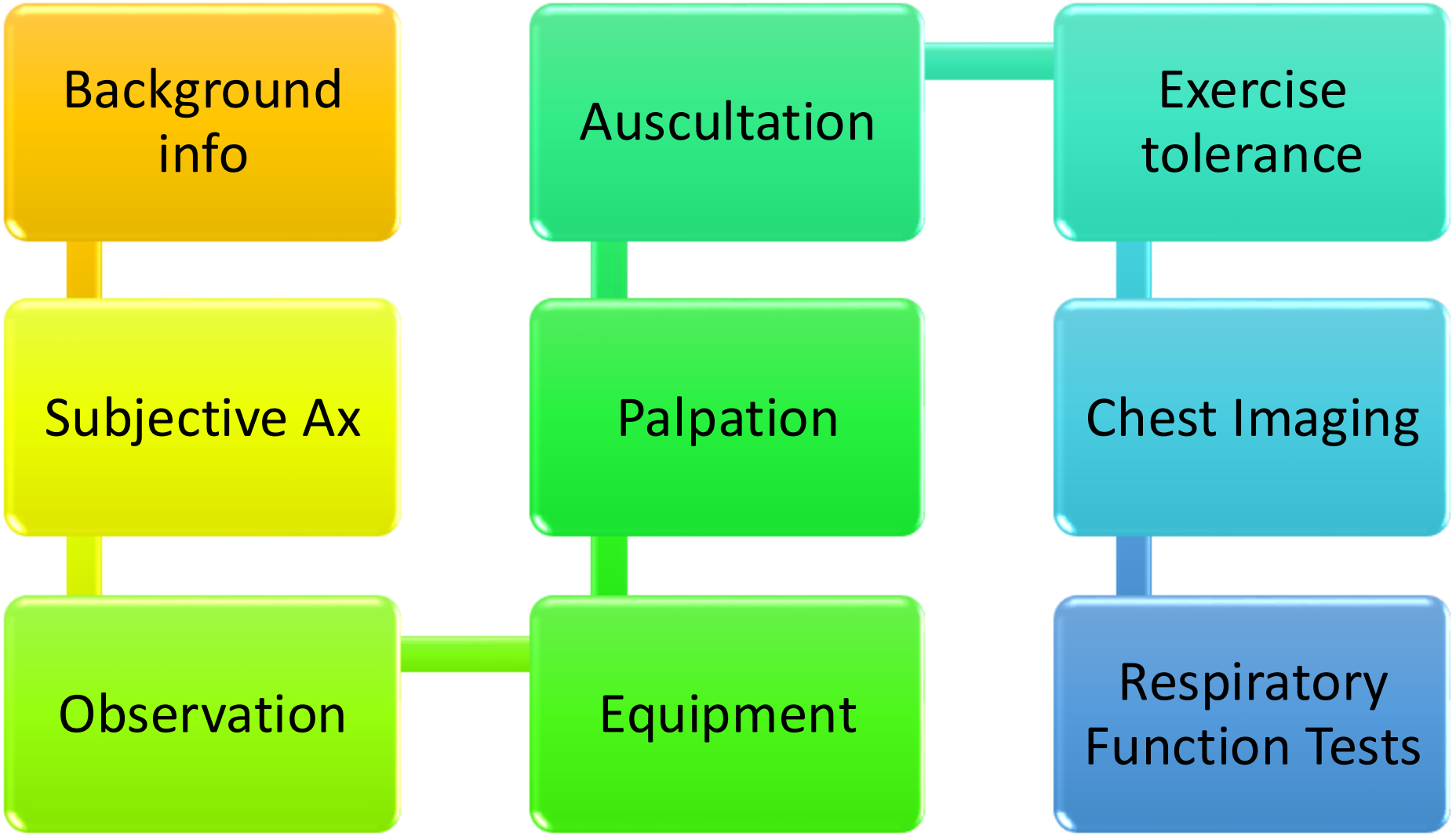




- Education
- Breathlessness management
- Airway clearance techniques
- Pulmonary Rehabilitation
- Inhaler review
- O2 management

Physiotherapy Management of COPD

Components of a cardiorespiratory Ax:



Subjective Assessment / Patient Interview Overview

Presenting Condition (PC)
& History of presenting condition

Past medical history (PMH)

Medications

Current Symptoms

Lifestyle Choices/
Smoking history

Exercise tolerance

Social history

Red flags

Other information

- Previous Physio/other input?
- As an inpatient/outpatient? Group/individual?
- Do they already have an ongoing treatment plan/self management plan?
- Vaccinations

Signs & Symptoms… What's the difference?

SIGNS = OBJECTIVE	• What we can assess / interpret / measure
SYMPTOMS= SUBJECTIVE	• What the patient is reporting to us • We generally record these first

SYMPTOMS

COUGH:
- Do you cough?
- Do you normally cough?
- What triggers it?
- Once you start, can you stop?
- Type of cough?
- Wet / dry / tickly?
- Productive?

PAIN:
- Do you have any pain?
- Where is it?
- How long have you had the pain?
- Is this a new pain or chronic?
- How severe is the pain?
- Aggs/Eases?

DYSPNOEA/SOB:
- Do you get short of breath?
- What activities make you short of breath?
- How would you rate your shortness of breath?
- What do you do when you are SOB?
- Have you had and recent changes in your SOB?
- Orthopnea (breathlessness when lying flat):Do you get short of breath when lying flat?
- How many pillows do you sleep on? Why? (may be for MSK issues no SOB)
- Paroxysmal nocturnal dyspnoea (PND):Do you wake in the night feeling breathless?

SPUTUM / SECRETIONS:
- Do you bring up sputum?
- Is that normal/new/different?
- Quality?
- Quantity?
- Smell?
- Easy / difficult to clear?
- Clearance techniques?

WHEEZE vs CHEST TIGHTNESS
- Do you feel wheezy? Do you feel tight chested?
- What brings it on?
- Does anything alleviate this? Do you use any medication to alleviate this?

OTHER:
- Dizziness
- Nausea
- Malaise
- Fevers/Chills
- Lethargy
- Headache

CARDIORESPIRATORY RED FLAGS:

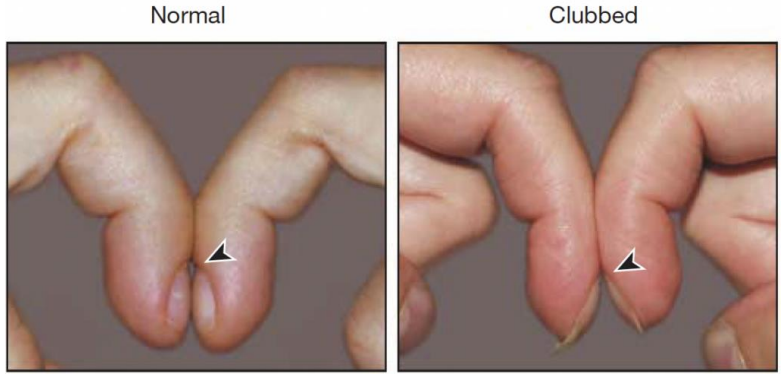


- Chest pain (explore further with **follow up questions**)
- Frank Haemoptysis / Pink frothy secretions
- Night sweats
- Uncontrollable Drowsiness / Daytime somnolence
- Unexplained weight loss / Significant weight gain within short period of time

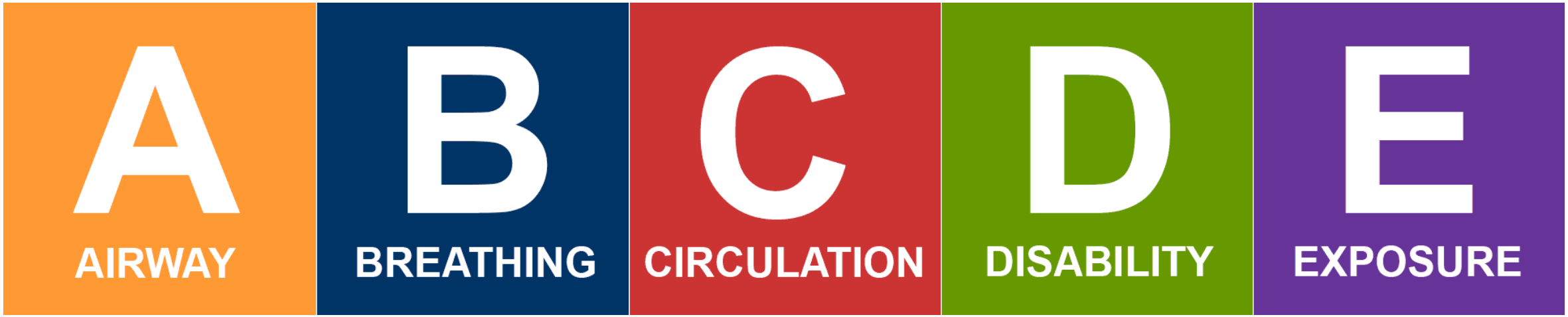


- \+ Temperature changes
- *Deranged Vital observations (High NEWS Score)*
- *Cyanosis / Mottled/ Non –blanching rash*
- *Clubbing*
- *Peripheral / Pitting oedema*
- *Confusion/Altered mental state*





Objective Assessment
ABCDE approach





Information regarding upper airway patency

- Look:
  - Own (self ventilating) / artificial airway?
  - Patency / Occlusion
- Feel:
  - Is air moving in and out of natural / artificial airway?
- Listen:
  - What can you hear to suggest a clear / obstructed airway?



Assessment of the anatomical and physiological features, contributing to the process of breathing

- Look:
  - Initial patient observations - position, ability to speak, movement etc
  - Mode of ventilation & FiO2 requirement
  - Attachments
  - Chest shape
  - Chest & abdominal movement
  - Work of breathing & breathing pattern
- Feel:
  - Palpation - Expansion, fremitus, subcutaneous air ?
- Listen:
  - Speech
  - Audible breath sounds
  - Auscultation
  - Cough
- Measure:
  - RR
  - SpO2 (& FiO2)
  - ABGs
  - Lung function tests
  - Peak cough flow
  - Respiratory muscle strength
  - Breathlessness
    - eg: Modified Bord, MRC Dyspnoea Scale, Dyspnoea 12
  - Thoracic Imaging

Palpation

Normal/Relaxed breathing:

Abdominal movement:

- Place one hand gently over the central area of the superior abdomen (just above belly button)
- Feel for in-drawing or diaphragmatic breathing

Thoracic movement:

- Basal: place both hands gently over the lateral chest wall – feel location, extent, symmetry
- Apical: place both hands (or one hand if patient significantly smaller than you) on anterior upper chest wall - inferior to clavicles, superior to breast.
    - Feel location, extent, symmetry, any fremitus.

Deep breathing:

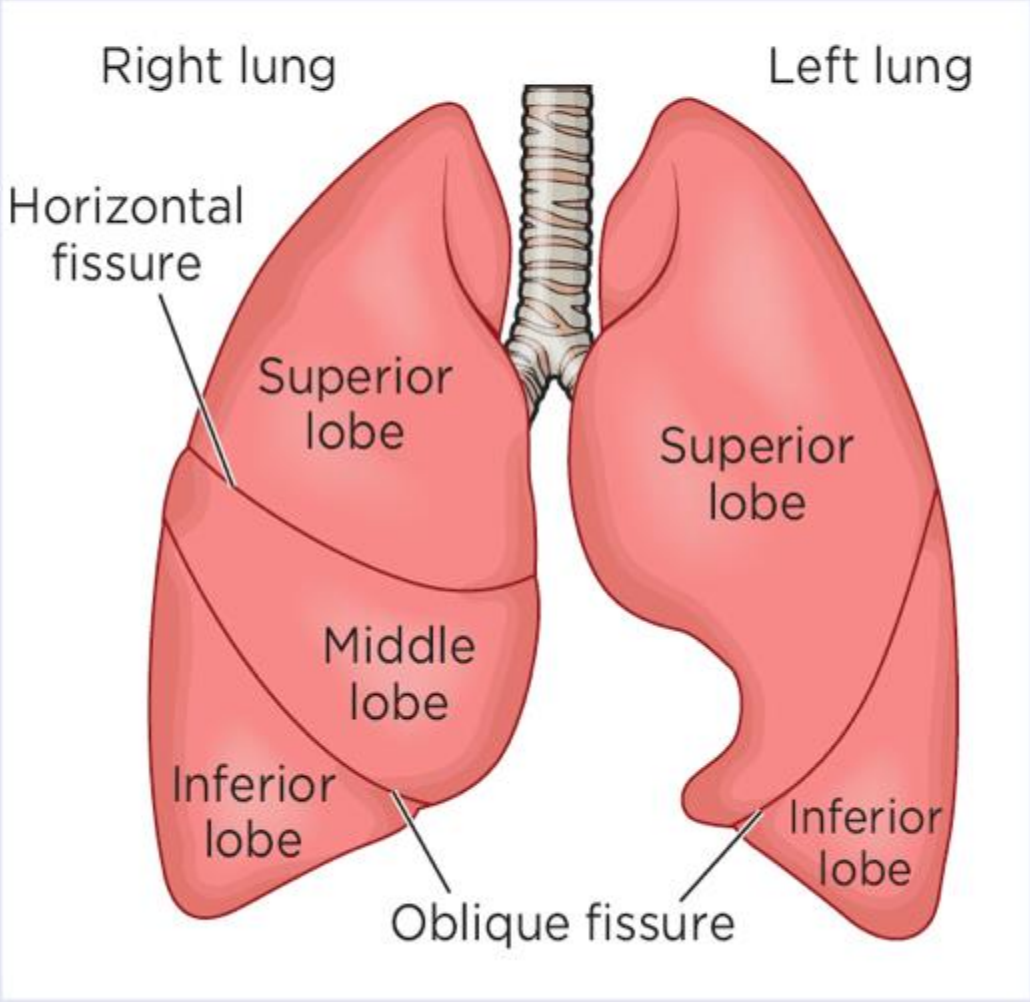
- Repeat same hand positions
- Especially feeling for thoracic movement



Auscultation:

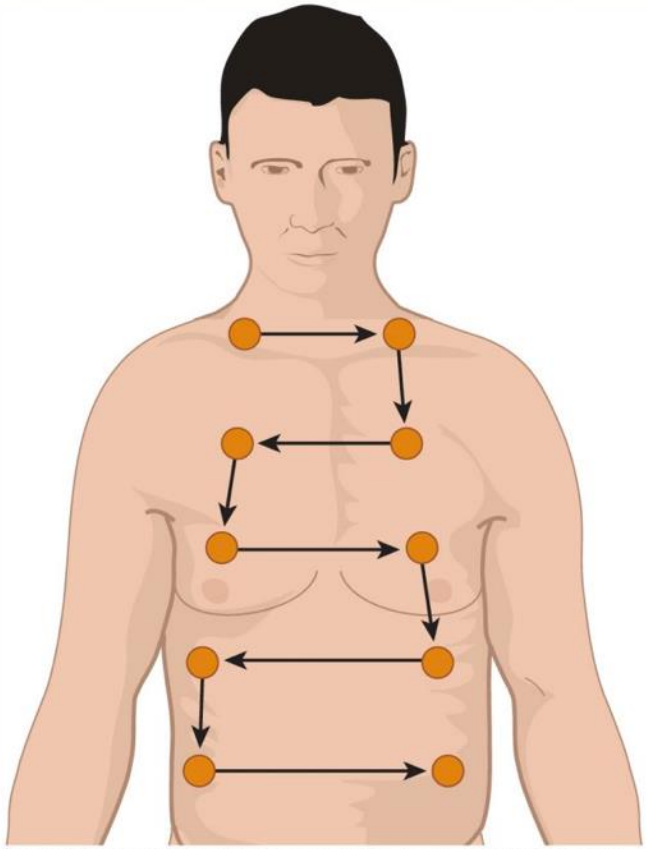
Positioning	Position patient appropriately • Able to access the anterior, posterior and lateral chest wall • Sitting upright / standing if possible • Slouch forward to move scapulae off the posterior chest wall
Breathing	Ask the patient to take slightly deeper breaths in and out through their mouth (generates less distortion/turbulence to airflow), at a rate a little faster than normal • Make sure you ask the patient if they need a rest / look out for signs of dizziness or fatigue.
Listen	Systematically listen to each lobe of the lungs • top to bottom • comparing right and left hand side • anterior & posterior • Listen to a full inspiration:expiration (I:E) cycle at each point

Fig 2. **Location of the lung lobes (anterior chest)**



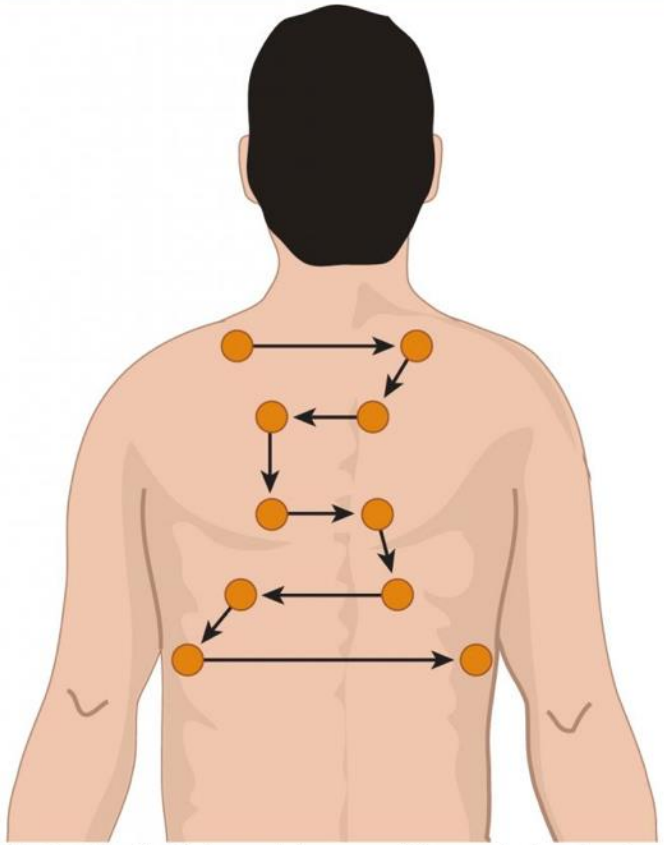
Auscultation:

Fig 4a. **Anterior chest auscultation**



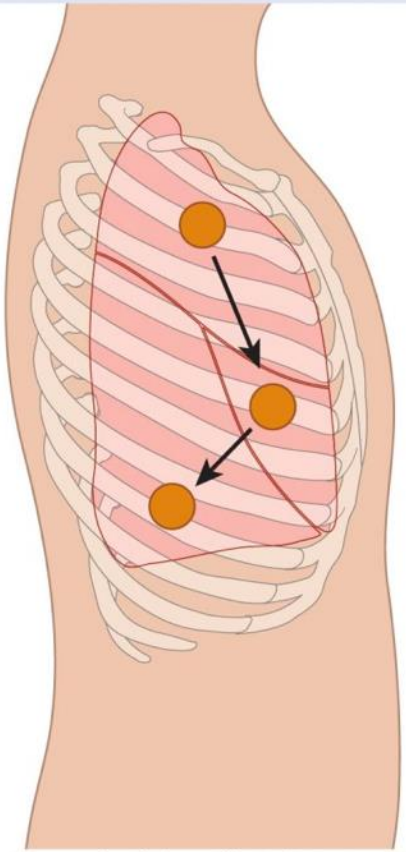
Starting at the top of the chest (first intercostal space), use a 'stepladder' approach to listen to breath sounds on the anterior chest finishing at the seventh intercostal space

Fig 4b. **Posterior chest auscultation**



Start at the first intercostal space of the posterior chest moving downwards, avoiding the scapula, to the seventh intercostal space

Fig 4c. **Right lateral chest auscultation**

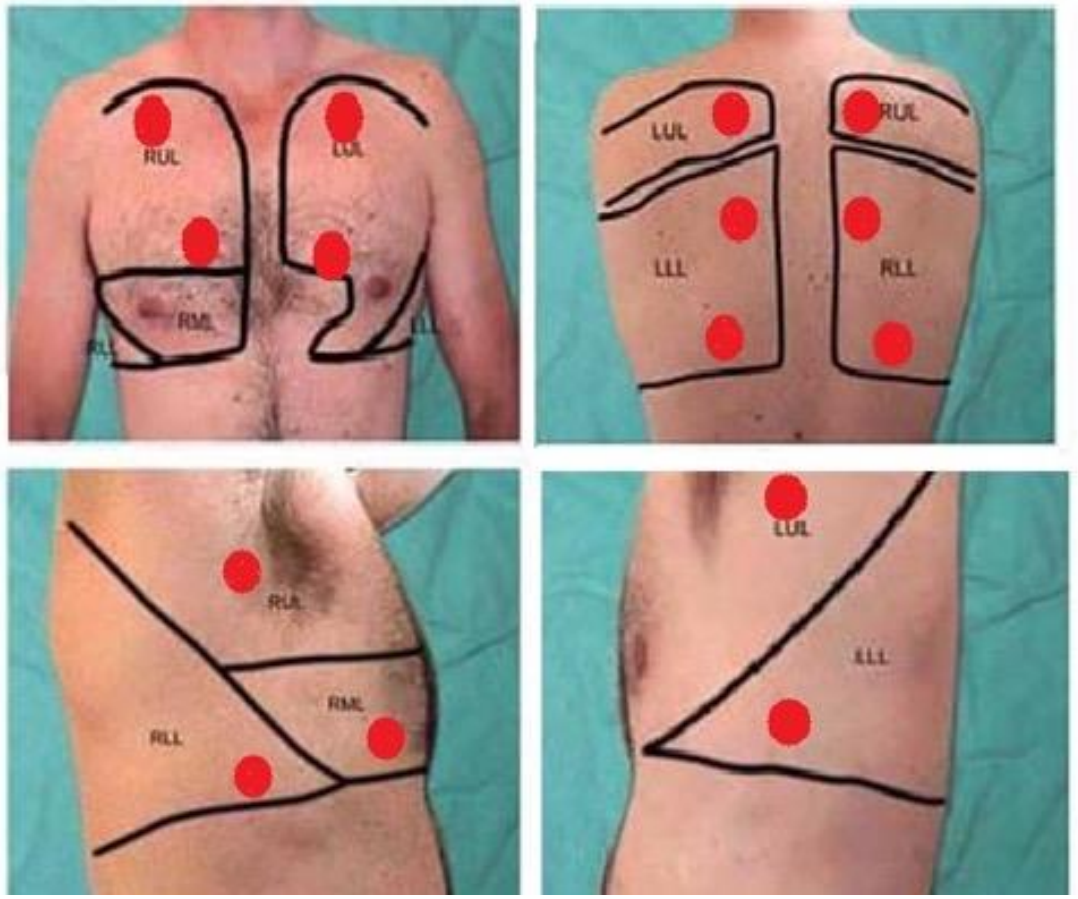


Move from the peak of the axilla to between the seventh or eight rib on the right and left.

Auscultation:

- Two questions you should ask yourself whilst listening:
  - **Are the breath sounds normal, increased or decreased?**
    - i.e. what is the volume like?
  - **Are there any added sounds?**

  - For the above two questions – ensure you note where you hear this
    - Listening anteriorly is predominantly upper lobes
    - Listening posteriorly is predominantly lower lobes.



Auscultation:

Timing

Inspiration/
Expiration

Pitch

High Pitch/

Low Pitch

Continuity

Discontinuous

or Continuous

Location of sound

Peripheral or

Larger airways

Defining characteristic

Harsh or squeaky or musical or snoring



Information regarding cardiovascular function:

- Look:
  - Complexion
  - Sweating
  - Moist / dry mouth
  - Cardiac support / monitoring
  - Lines
  - Skin turgor
  - Capillary refill time
  - Peripheral oedema
  - Urine colour / quality
- Feel:
  - Temperature
  - Peripheral pulses
  - Pulse rate
  - Pulse rhythm & quality
- Listen:
  - Heart sounds – n/a
- Measure:
  - Temperature
  - ECG
  - BP
  - Central venous pressure
  - Fluid balance
  - Cardiac Output (CO)
  - Cardiac Index (CI)
  - Pulmonary artery occlusion pressure (PAOP)
  - Exercise testing



Information regarding neuromusculoskeletal function:

- Look:
  - Glasgow Coma Scale
  - AVPU
  - Sedation score (RASS)
  - Pupillary response
- Feel:
  - Resistance – nature of
  - Rigidity
  - Spasticity
- Listen:
  - Speech
  - Confusion / disorientation
  - Dysarthria
  - Aphasia
  - Dysphagia
- Measure:
  - Full neuro ax required?
  - Sensation
  - Muscle strength
  - ROM
  - Functional ability
  - Blood glucose
  - ICP
  - QoL

National Early Warning Score (NEWS) 2

Physiological parameter	Score 3	2	1	0	1	2	3
Respiration rate (per minute)	≤8		9–11	12–20		21–24	≥25
SpO_2 Scale 1(%)	≤91	92–93	94–95	≥96			
SpO_2 Scale 2(%)	≤83	84–85	86–87	88–92 ≥93 on air	93–94 on oxygen	95–96 on oxygen	≥97 on oxygen
Air or oxygen?		Oxygen		Air			
Systolic blood pressure (mmHg)	≤90	91–100	101–110	111–219			≥220
Pulse (per minute)	≤40		41–50	51–90	91–110	111–130	≥131
Consciousness				Alert			CVPU
Temperature (°C)	≤35.0		35.1–36.0	36.1–38.0	38.1–39.0	≥39.1	

Behaviour	Response
 Eye Opening Response	4. Spontaneously 3. To speech 2. To pain 1. No response
 Verbal Response	5. Oriented to time, person and place 4. Confused 3. Inappropriate words 2. Incomprehensible sounds 1. No response
 Motor Response	6. Obeys command 5. Moves to localised pain 4. Flex to withdraw from pain 3. Abnormal flexion 2. Abnormal extension 1. No response

Awake	Patient is awake
Verbal	Patient responds to a verbal stimulus
Pain	Patient responds to a pain stimulus
Unresponsive	Patient is unresponsive to stimulus



Other physical features / any other relevant information:

- Look:
  - Injuries
  - Wounds
  - Attachments

- Measure:
  - Weight
  - Blood tests

Other Components

Exercise Capacity Testing

- Shuttle walk / 6MWT
- Oxygen saturations during exercise

Function

How well do they:

- Walk
- Complete transfers
- Get out of bed
- Management of ADLS

MSK/ Neurological

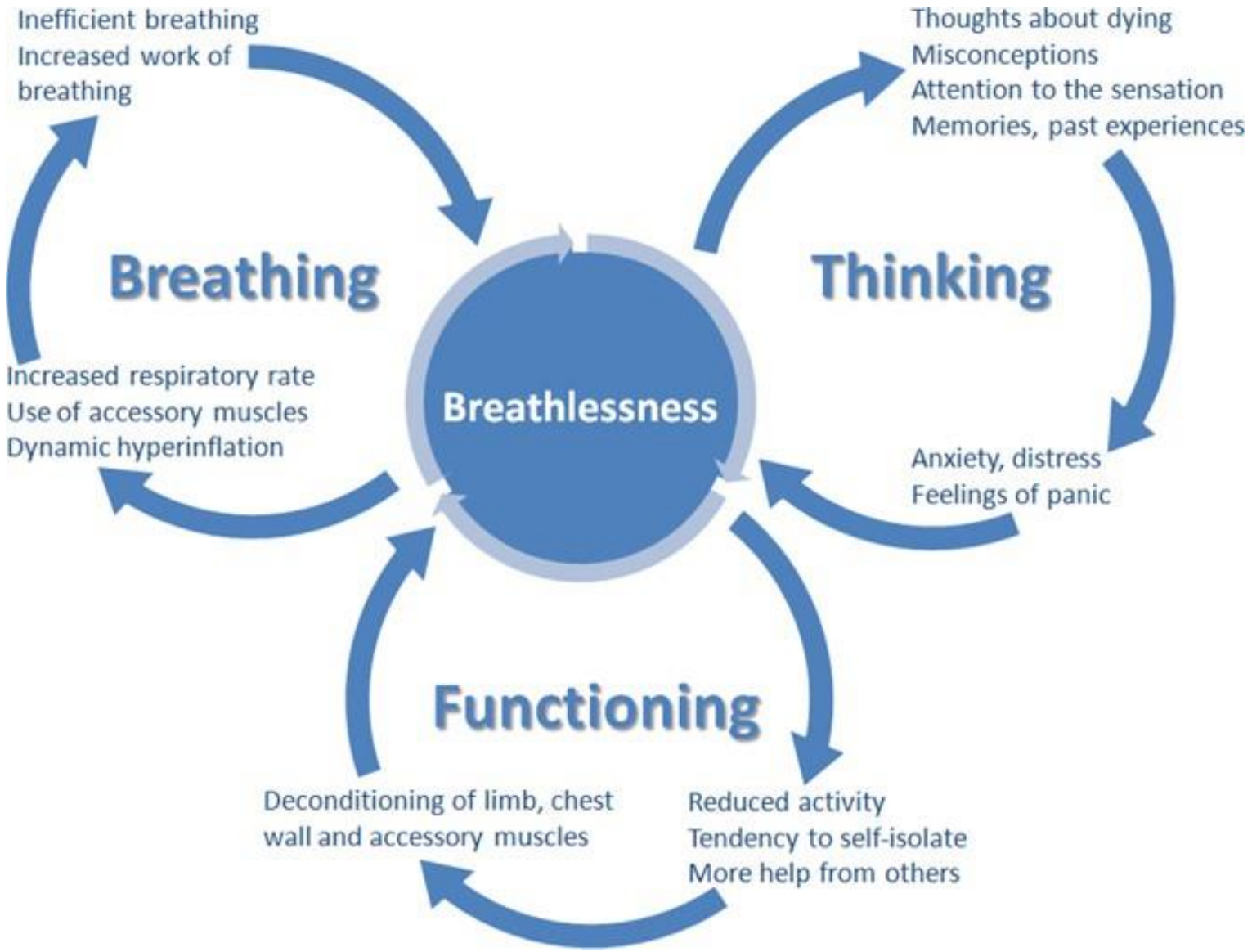
- Muscle strength
- Sensation / proprioception
- Balance

CONSENT

Always explain the process to the patient and gain consent before you move forward with performing any element of the physical exam.



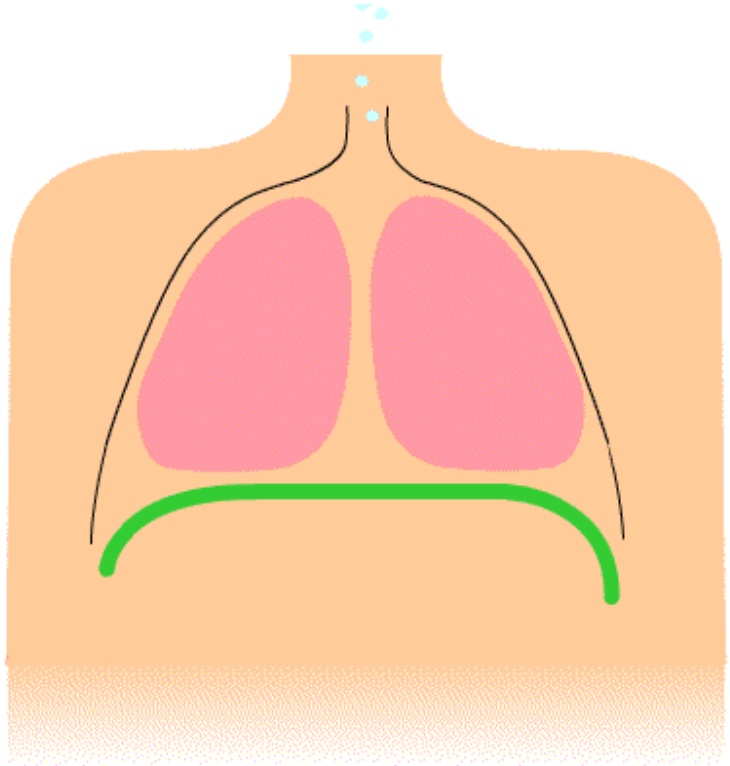
Management of Breathlessness



(Spathis *et al.,* 2017)

Positioning to Relieve Breathlessness

- Leaning forward in standing or sitting moves abdominal contents so that the anterior part of the diaphragm is raised
- Similar effect achieved in-side lying/high side lying
- Increased curvature of diaphragm
- Optimises length tension relationship of the diaphragm
- Facilitates diaphragmatic contraction and reduces work of breathing



Breathing Techniques

Breathing Control/Diaphragmatic Breathing:

- Reduces WOB
- Reduces RR
- Reduced energy expenditure
- Decreased oxygen demand

Pursed Lip Breathing:

- Creates positive expiratory pressure
- Prevents airway collapse
- Improved lung compliance
- Ensure neck and shoulder muscles are relaxed
- Inhale through the nose and gently out through pursed lips

Blow-as-you-go:

- Breathing out during the effortful part of any task e.g. ascending stairs
- Avoids breath holding/inhaling with effort – valsalva maneuver
- Use pursed lips
- Can be applied to any activities/exercise

Principles of Exercise in Cardiorespiratory Disease:

High prevalence of comorbidity in people with respiratory disease that is influenceable with exercise

Offsetting of complications associated with respiratory medications

Modulates elements of the disease itself

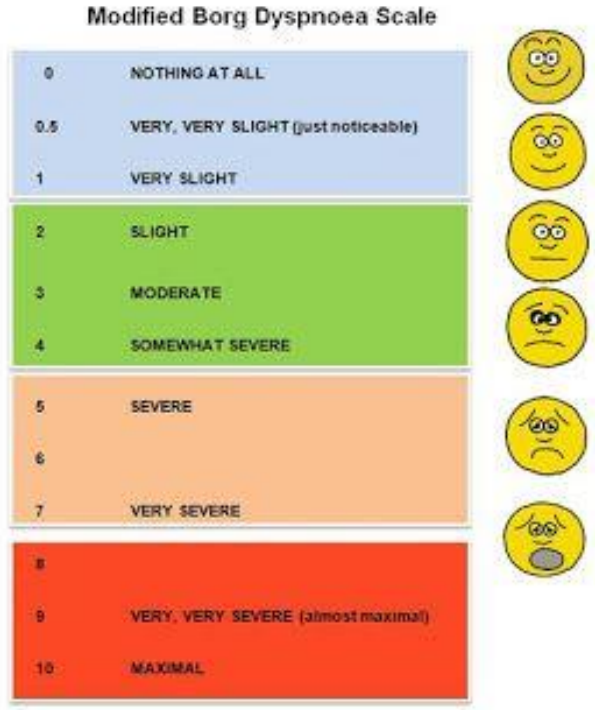
- Evidences support that exercise is able to modulate inflammatory pathways in asthma and COPD, including cytokines and leukotriene pathways
- Reversal/slow down cachexia in interstitial lung disease
- Strengthens immune response therefore reduce infection risks

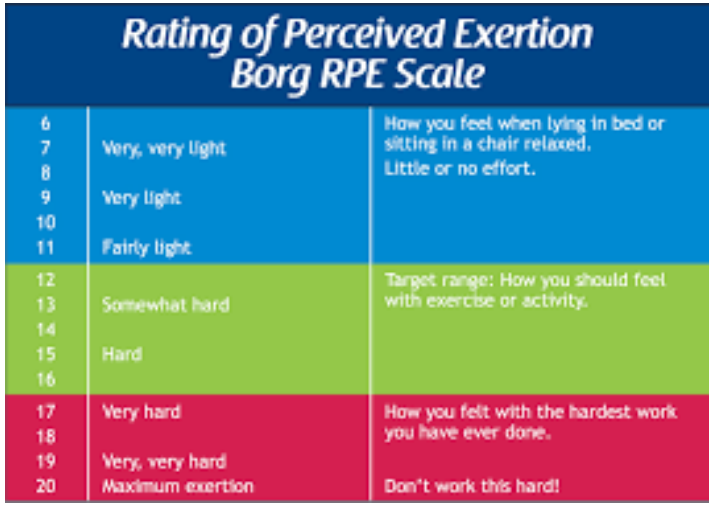
Pacing progression

- Want to avoid boom and bust
- Avoid discouraging- Too much too soon
- Be wary of the patient group- Not all will have exercised before
- We want exercise to be a habit for life not a quick fix
- Avoidance of post exertional malaise

MRC DYSPNOEA SCALE (for COPD patients)	
Grade	**Impact**
1	Not troubled by breathlessness except on vigorous exertion
2	Short of breath when hurrying or walking up inclines
3	Walks slower than contemporaries because of breathlessness, or has to stop for breath when walking at own pace
4	Stops for breath after walking about 100 m or stops after a few minutes' walking on the level
5	Too breathless to leave the house or breathless on dressing or undressing

Grade	Description of Breathlessness
Grade 0	I only get breathless with strenuous exercise
Grade 1	I get short of breath when hurrying on level ground or walking up a slight hill
Grade 2	On level ground, I walk slower than people of the same age because of breathlessness, or I have to stop for breath when walking at my own pace on the level
Grade 3	I stop for breath after walking about 100 yards or after a few minutes on level ground
Grade 4	I am too breathless to leave the house or I am breathless when dressing







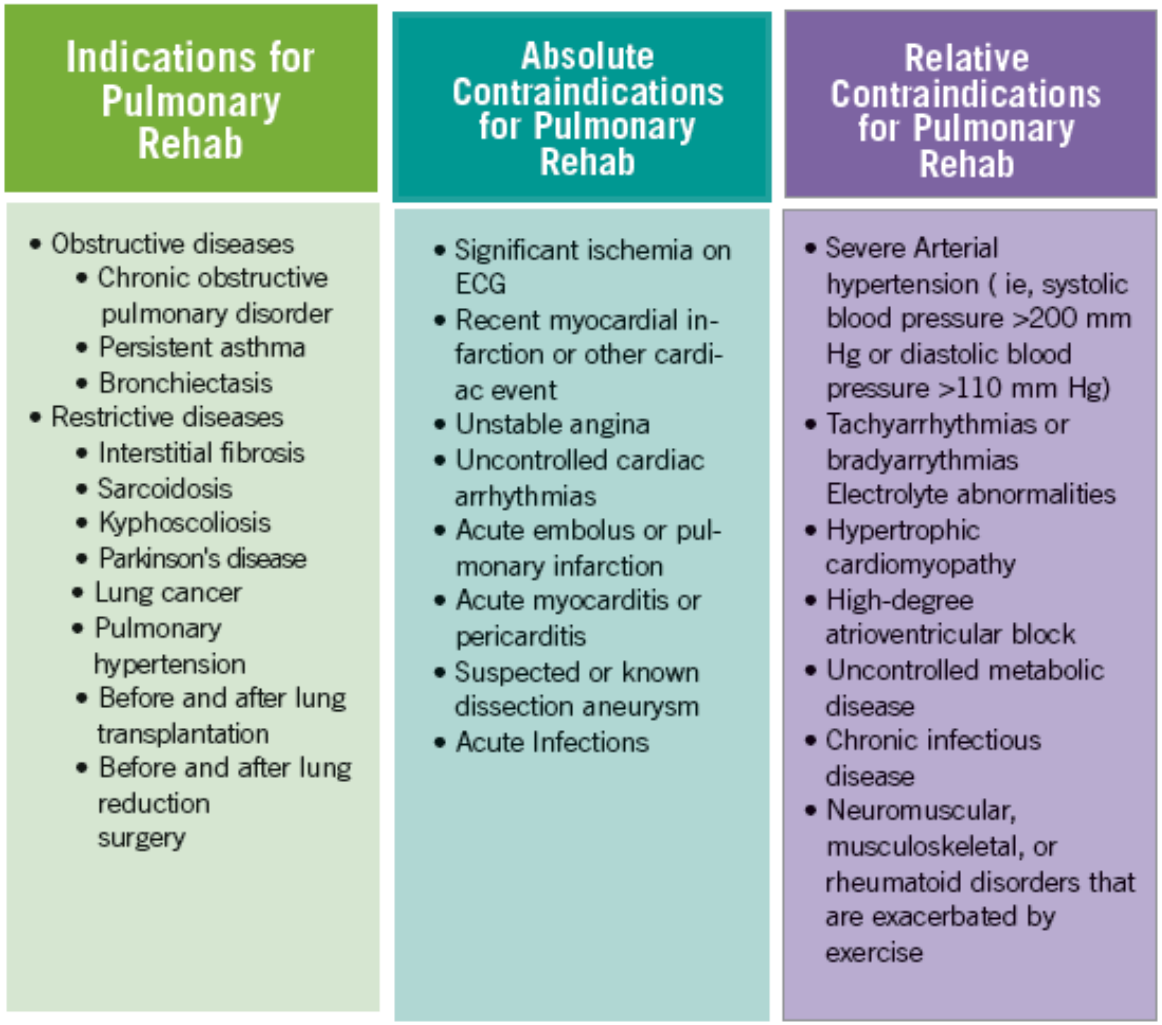
Benefits of Pulmonary Rehab

- ✓ Increased exercise capacity
- ✓ Reduced dyspnoea
- ✓ Improved health status
- ✓ Improved physical activity
- ✓ Improved self-reported measures of ADL
- ✓ Increased muscle strength
- ✓ Reduced anxiety and depression, improved general psychological well being
- ✓ Cost effective

(BTS Guidelines, 2013)

PR Inclusion/Exclusion Criteria:

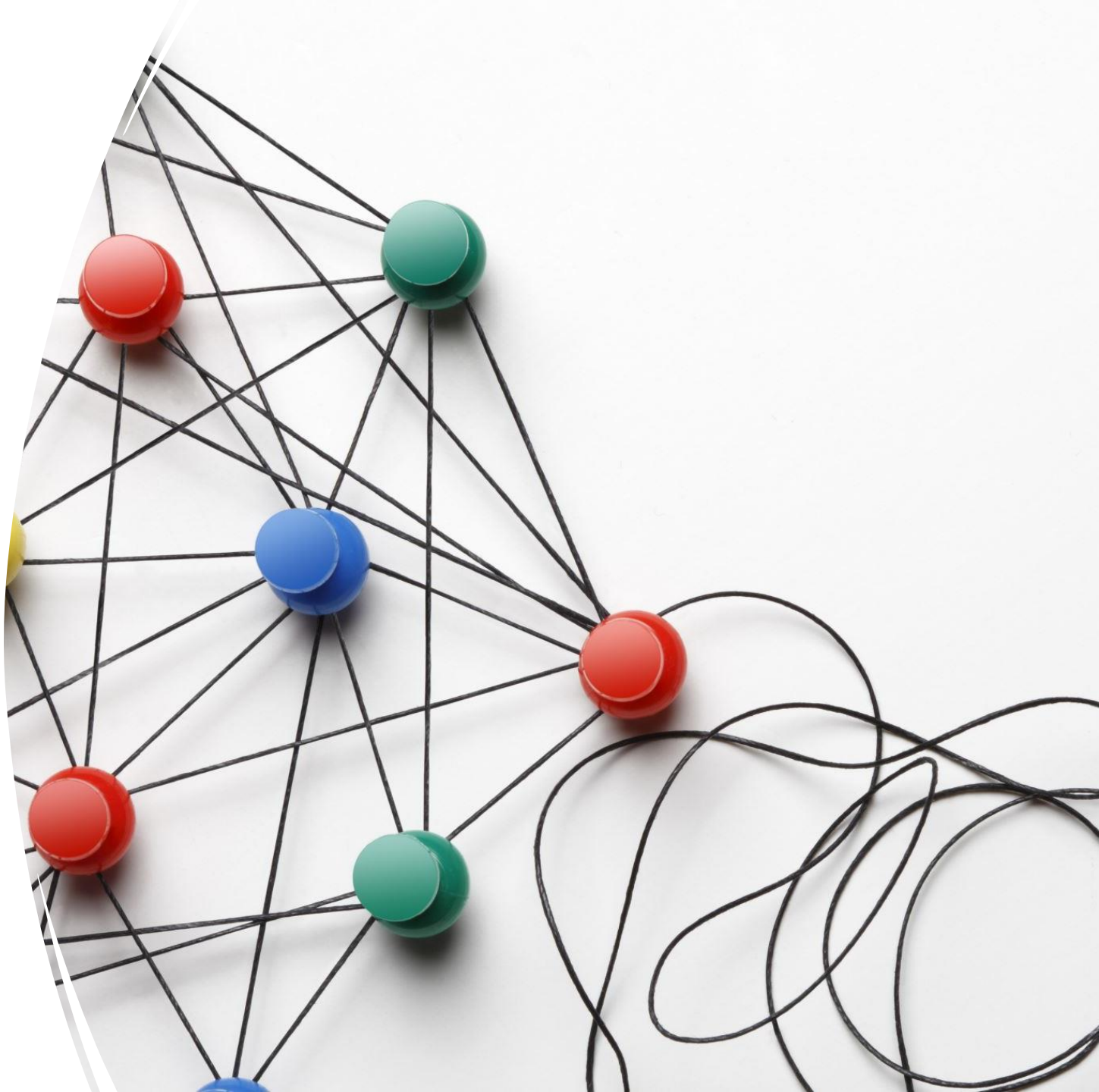
- **COPD**
  - Functionally limited by SOB (MRC 2-5)
  - People admitted to hospital for an acute exacerbation of chronic obstructive pulmonary disease (COPD) should start a pulmonary rehabilitation programme within 4 weeks of discharge.
- **Current smokers**
  - Can refer to smoking cessation services
- **Other respiratory conditions**
  - Non-CF bronchiectasis
  - Interstitial lung disease
  - Asthma (with fixed airflow limitation)



Exercise Tolerance Testing:

Appropriate methods include:

- Incremental shuttle walk test (ISWT)
- Endurance shuttle walk test (ESWT)
- Six-minute walk test (6MWT)
- 1 Rep Max for Strength based component
- 4 meter gait speed
    - MCID (Minimum Clinically Important Difference) is an improvement in speed > 0.11m/sec.
- 5 x Sit to Stand (FTSTS)
    - MCID is an improvement is a reduction in time >1.7Seconds





Example Class Structure:

- **Exercise component:**
  - Warm up - 10 mins
  - Aerobic exercises - 20 mins total
    - 1 minute break after first 10 mins
    - Second 10 mins completed as able
  - Strengthening exercises
    - Patients work at own pace
    - Sitting/standing
    - Aiming for 3 sets of 10 reps per exercise
  - Cool down – 10 mins
- **Post-exercise assessment**
- **Education session**

QUESTIONS?

